

Overview

A high level of protection is given to wildlife, plants, trees and hedgerows through legal controls and contract conditions.

This chapter gives a general overview of the legal framework for the protection of wildlife and their habitats. It highlights endangered and protected species (animals and plants) and designated sites for their protection.

This chapter also gives an overview of the legal framework for the regulation of invasive species.

5.1 Introduction

Ecology refers to the relationships between plants, animals, people and their environments, and to the interaction within these relationships. An ecosystem is made up of plants, animals, micro-organisms, soil, rocks, minerals, water sources and the local atmosphere interacting as well as sustaining one another over a certain period of time.

The identification and management of wildlife needs to be undertaken early in the planning stage of a project to avoid costly delays to the programme and possible loss of reputation if damage takes place. Damaging, disturbing or removing protected species can result in prosecution under a range of environmental legislation, and wildlife is also held in high regard by the public.

Construction work (such as demolition, site clearance and dewatering) potentially impacts plants and wildlife in the following ways.

- Removal and fragmentation of habitats.
- Disturbance to aquatic wildlife and water quality.
- Disturbance to wildlife from noise and vibration.
- Damage to trees and hedgerows.
- Changes in lighting conditions.



Refer to Appendix C for a list of construction work and its potential adverse effects on wildlife.

When making a planning application, developers must produce an ecological impact assessment identifying intended mitigation measures before, during and after construction work.

5.2 Important points



The following practices should be employed to avoid damage to wildlife and its habitats:

Plan in order to mitigate for ecological issues, before the work starts

- Employ a qualified ecologist to carry out a preliminary ecological appraisal (also known as Phase 1 habitat survey). The appraisal will identify wildlife features and ecologically important areas prior to works commencing and designate them as protected areas (for example, fence them off). The contract documentation should identify sensitive areas that will require protection and management. The appraisal will:
 - gather baseline ecological data using Phase 1 habitat survey guidelines (JNCC 2010)
 - make recommendations for detailed surveys, where required
 - identify key constraints to the project and make recommendations for design options to avoid significant impact on important ecological features
 - identify any necessary mitigation measures
 - identify enhancement opportunities
 - aid site management plans by mapping out ecological features
 - inform any BREEAM (land use and ecology credits) reports.
- Liaise with statutory bodies and local groups to explain any mitigation measures to be used.
- Where there is a need to take, disturb or relocate protected species, consents should be obtained from the relevant regulatory body and competent licensed ecologists used to carry out the work (*refer to 5.5*).
- Plan site clearance or demolition works to avoid any nesting, hibernation or breeding seasons (*refer to Appendix A for a yearly ecology planner that identifies the constraints and best times for dealing with the main groups of protected species*).
- Inform site personnel of any protected areas and explain the consequences of any damage to these areas.



The following practices should be employed to avoid damage to wildlife and its habitats (continued):

During construction

- Regularly check the condition of fencing of any designated protected areas.
- Refer to the ecology year planner (*Appendix A*) to be aware of differing seasons and constraints.
- Ensure watercourses are free from contaminated run-off or any other forms of pollution.
- Confirm compliance against method statements, the construction environmental management plan and any environmental contractual requirements.
- In the event of any unexpected ecological finds, stop work and then consult with the site ecologist (if relevant) and statutory bodies.

5.3 Designated sites

The UK has a responsibility to ensure the protection of species and their habitats from both a national and international perspective. One approach to achieving this is to establish designated sites that are legally protected, as detailed below.

Areas of Outstanding Natural Beauty (AONBs). In England, Northern Ireland and Wales, the primary purpose of the AONB designation is to conserve natural beauty. This, by statute, includes wildlife, physiographic features and cultural heritage, as well as the more conventional concepts of landscape and scenery. Account is taken of the need to safeguard agriculture, forestry and other rural industries and the economic and social needs of local communities. AONBs have equivalent status to National Parks as far as conservation is concerned.

AONBs are designated under the National Parks and Access to the Countryside Act 1949 amended in the Environment Act 2021. The Countryside and Rights of Way Act 2000 clarifies the procedure and purpose of designating AONBs. Originally designated in Northern Ireland under the Amenity Lands Act (Northern Ireland) 1965, AONBs are now designated under the Nature Conservation and Amenity Lands (Northern Ireland) Order 1985. In Scotland, National Scenic Areas are broadly equivalent to AONBs.

Areas of Special Protection for birds (AOSPs) (England and Wales) and **wildlife refuges** (Northern Ireland) are established under the Wildlife and Countryside (Amendment) Act 1991 and the Wildlife (Amendment) (Northern Ireland) Order 1995. The purpose of such orders is normally to provide sanctuary to particularly vulnerable groups of birds. In Scotland, these areas are called Special Protection Areas (SPAs).

Biogenetic reserve. A number of National Nature Reserves (NNRs) and some important Sites of Special Scientific Interest (SSSIs) have been identified as biogenetic reserves under a Council of Europe programme for the conservation of heathlands and dry grasslands.

Country parks are statutorily declared and managed by Local Authorities in England and Wales under the Wildlife and Countryside (Amendment) Act and in Scotland under the Countryside (Scotland) Act 1967. In Northern Ireland they exist as a non-statutory designation. Country parks are primarily intended for recreation and leisure opportunities close to population centres and do not necessarily have any nature conservation importance. Nevertheless, many are in areas of semi-natural habitat and so form a valuable network of locations at which informal recreation and the natural environment coexist.

Local Nature Reserves (LNRs) (England, Scotland and Wales) and **Local Authority Nature Reserves (LANRs)** (Northern Ireland) may be established by Local Authorities under Section 21 of the National Parks and Access to the Countryside Act. These habitats of local significance can make a useful contribution to nature conservation and public amenity.

Marine Nature Reserves (MNRs) are designated under the Wildlife and Countryside (Amendment) Act (England, Scotland and Wales) to conserve marine flora or fauna, geological or physiographical features, or to allow study of such features. In Northern Ireland they are designated under the Nature Conservation and Amenity Lands (Northern Ireland) Order 1985.

National Nature Reserves (NNRs) are areas of national, and sometimes international, importance that are owned, leased or approved by the relevant regulatory authority (such as Natural England or Natural Resources Wales), or are managed in accordance with nature reserve agreements between the regulatory authorities and landowners and occupiers. The essential characteristic of NNRs is that they are primarily used for nature conservation.

Ramsar sites. The Ramsar convention requires the protection of wetlands that are of international importance, particularly as waterfowl habitats.

Sites of Special Scientific Interest (SSSIs) (England, Scotland and Wales) and **Areas of Special Scientific Interest (ASSIs)** (Northern Ireland). All sites of national and international importance on land, including NNRs, Nature Conservation Review (NCR) sites, Geological Conservation Review (GCR) sites (England, Scotland and Wales), Earth Science Conservation Review (ESCR) sites (Northern Ireland), Special Protection Areas (SPAs), Special Areas of Conservation (SACs) and Ramsar sites, are notified as SSSIs (England, Scotland and Wales) or ASSIs (Northern Ireland).

Owners and occupiers are required to notify the relevant regulatory authority of potentially damaging operations and may not undertake them for four months unless they are in accordance with the terms of a management agreement with consent of the relevant regulatory authority.

As protection, the Secretary of State may make a nature conservation order to protect any sites of national or international importance. SSSIs are classed as such under the Wildlife and Countryside (Amendment) Act. In Northern Ireland the law relating to ASSIs is contained in the Environment (Northern Ireland) Order 2002.

Local planning authorities are required to consult with the statutory bodies (Natural England, Natural Resources Wales, NatureScot and Northern Ireland Environment Agency) prior to allowing any development to proceed that may affect an SSSI or ASSI. Water, gas and electricity companies must also do the same.

Special Protection Areas (SPAs) and **Special Areas of Conservation (SACs)** are intended to protect the habitats of threatened species of wildlife. SACs are strictly protected sites under the EC Habitats Directive. This states the requirement to establish a European network of important, high-quality conservation sites. SPAs are strictly protected sites classified in accordance with Article 4 of the EC Birds Directive on the conservation of wild birds (79/409/EEC). They are classified for rare and vulnerable birds, listed in Annex I to the Birds Directive, and for regularly occurring migratory species.

5.4 Promoting biodiversity



Biodiversity is defined as the variety of plant and animal life in the world or in a particular region or ecosystem. A high level of biodiversity is considered to be important and desirable for the survival of ecosystems and species.

The objective of the Conservation of Habitats and Species Regulations 2017 is to protect biodiversity. The regulations ensure that the deliberate introduction into the wild of any species which is not native to their territory is regulated so it does not negatively impact natural habitats within their natural range or the wild native fauna and flora and, if necessary, prohibit such introduction.

Working together and following good design principles, planners, developers and clients should aim to minimise the impact of the project on ecological factors (such as habitat destruction, fragmentation and species isolation). Opportunities to promote biodiversity and support ecosystem services should be identified and, wherever possible, should actively seek to rebuild local ecological networks. This can be achieved through the actions set out below.

Enhance the overall ecological quality of the site and the surrounding ecological network by creating new habitats, buffer areas and landscape features that are of importance for wildlife. Focus should be made:

- in areas where the most important, fragile and/or threatened habitats and species are known to occur
- where there are species requiring large ranges (land areas required for them to exist), which have particularly suffered as a result of their habitats becoming reduced in size and isolated
- on species with low reproductive capacity (most large mammals), species highly sensitive to disturbance (most birds of prey) and species subject to recovery programmes (the focus of local biodiversity action plan targets).

Local biodiversity action plans, species action plans and habitat action plans should be used as a guide to the relevant priorities for such positive measures at the local level. These may include rehabilitation of degraded habitats or the creation of new habitats within and adjacent to development sites. Bird boxes, swift bricks, bat boxes, bat bricks, green roofs, green walls and habitat creation are examples where enhancement can be achieved on construction projects.

Avoid developing sites, and locations within sites, where existing habitats, important species, buffer areas and other landscape features of major importance for wildlife would:

- suffer direct impact resulting in the reduction or complete loss of habitat and/or diversity of species present
- suffer an indirect impact from nearby development through increased ecological disturbance and stress, thereby reducing the site's capacity to support the wildlife present
- suffer a reduction in ecological quality, so that the site is unable to support migration, dispersal or genetic exchange of wild species
- be further fragmented from other similar features by development that causes a barrier effect in the landscape between fragments.

Restore and, where possible, link and connect existing habitats and landscape features, which could potentially be of major importance for wildlife, enhancing their ability to support migration, dispersal and genetic exchange.

Retain and incorporate within the development site existing habitats, important species, buffer areas and landscape features of major importance for wildlife – making sure that the site retains at least the same capacity to support the diversity, abundance, migration, dispersal and genetic exchange of wildlife as it did prior to development.

Compensate for features lost to development through the following.

- Re-creation, as nearby as possible, of features and landforms capable of maintaining the same quality of habitats and species as would otherwise be lost or displaced through the development.
- Restoration and enhancement of surrounding or nearby features unaffected by development.
- Creation of new or additional buffer areas to reduce impacts.
- Translocation of habitats and species that would otherwise be lost.



For further information on local biodiversity action plans visit the Joint Nature Conservation Committee (JNCC) website.

For practical advice visit the CIRIA website for its *Working with wildlife* mobile app.

5.5 Regulatory bodies for nature conservation

Organisations with responsibility for nature conservation in each of the devolved administrations are shown in the table below.

Area	Responsible organisation
England	Natural England
Northern Ireland	Northern Ireland Environment Agency
Scotland	NatureScot
Wales	Natural Resources Wales

The regulatory bodies' remit includes providing ecological advice, consultation during the planning process, promoting biodiversity, and protecting designated ecological sites and protected species. The Joint Nature and Conservation Committee (JNCC) is the main public body that advises the UK Government and devolved administrations on UK-wide and international nature conservation.

JNCC is led by the Joint Committee, which brings together members from the nature conservation bodies for England, Northern Ireland, Scotland and Wales and independent members appointed by the Secretary of State for the Environment, Food and Rural Affairs under an independent chairperson. There are many other organisations, national and regional, involved in the protection and enhancement of plants and wildlife across the UK. Below are a few examples of the more prominent organisations.

- The National Trust.
- The Wildlife Trust.
- The Royal Society for the Protection of Birds.
- National parks.

5.6 Endangered species

Nature is humanity's life support system, and underpins all life on Earth. The natural world slowly changes over thousands of years, during which time new species evolve and some species decline and may ultimately become extinct. Human activity can have a dramatic impact on these natural cycles by putting significant pressure on ecosystems through vegetation clearance, deforestation and habitat fragmentation caused by agricultural development, road building and other infrastructure projects. This can lead to an accelerated decline in species and the habitats that support them.

The responsible management of wildlife and its habitat is critical if we want to maintain a healthy, sustainable planet and ensure its renewable resources are passed on to future generations.

At an international level, the International Union for Conservation of Nature (IUCN) is recognised for its work in publishing the *Red list of threatened species*, which is the definitive international standard for species extinction risk. The *Red list* is regularly updated with new information about the status of endangered species. Estimates based on the IUCN Red list indicate that one million species are currently threatened with extinction. However, it is not too late to reverse these trends.

In the UK, priority species and habitats are those that have been identified as being the most threatened and requiring conservation action. Section 41 of the Natural Environment and Rural Communities Act identifies priority species and habitats at risk of human-induced extinction.

COP 15, the international biodiversity conference held under the United Nations Convention on Biological Diversity (CBD) implemented a new set of international goals for biodiversity known as the Kunming-Montreal Global Biodiversity Framework (GBF). 188 governments, including the UK, agreed to the GBF and committed to addressing the ongoing loss of terrestrial and marine biodiversity. The GBF contains four main goals and 23 targets.

The four global goals set out a vision to address ongoing biodiversity loss by 2050:

Goal A. Substantially increase the area of natural ecosystems by maintaining, enhancing or restoring the integrity, connectivity and resilience of all ecosystems. Reduce by tenfold the extinction rate and risk of all species and increase the abundance of native wild species. Maintain the genetic diversity of wild and domesticated species and safeguard their adaptive potential.

Goal B. Ensure nature's contributions to people are valued, maintained and enhanced, with those contributions currently in decline being restored.

Goal C. Share the monetary and non-monetary benefits of the utilisation of genetic resources, digital sequence information on genetic resources, and traditional knowledge associated with genetic resources with Indigenous people and local communities. Additionally, ensure traditional knowledge associated with genetic resources is appropriately protected.

Goal D. Ensure all parties (specifically developing countries) have adequate means to implement the GBF. This includes financial resources, capacity building, technical and scientific cooperation, and access to technology.

The 23 global targets for 2030 include:

- Effective conservation and management of at least 30% of the world's lands, inland waters, coastal areas and oceans, with emphasis on areas of particular importance for biodiversity and ecosystem functioning and services. This has been informally referred to as the '30 by 30' deal.
- Reduce global food waste by half and significantly reduce over-consumption and waste generation.

- Reduce by half both excess nutrients and the overall risk posed by pesticides and highly hazardous chemicals.
- Progressively phase out or reform subsidies that harm biodiversity by at least \$500 billion per year, whilst scaling up positive incentives for biodiversity conservation.
- Encourage at least \$200 billion per year in domestic and international biodiversity-related funding from public and private sources.
- Increase international financial flows from developed to developing countries by at least \$20 billion per year by 2025, and to at least \$30 billion per year by 2030.
- Require transnational companies and financial institutions to monitor, assess and disclose the impact on biodiversity of their operations, supply chains and portfolios.

The GBF is not legally binding, but it does require countries to monitor and report their progress against the goals and targets every five years, or earlier. The 2023 update of the UK Biodiversity Indicators, published in November 2023, comprises updates to 18 indicators. Full information for each individual indicator, including charts, assessments and access to background information and resources including data and technical background documents, is available online.

[www](#) For more information, visit the Government website.

The UK lists of priority species and habitats remain important and are valuable reference sources. They were used to help draw up statutory lists of priorities in England, Northern Ireland, Scotland and Wales. These UK priority species were identified as being the most threatened and requiring conservation action. The list of UK priority species and habitats represents the most comprehensive analysis of such information ever undertaken in the UK, and now contains **1,149 species** and **67 habitats**. Species were assessed according to four criteria, as shown below.

- Threatened internationally.
- International responsibility and a 25% decline in the UK.
- More than 50% decline in the UK.
- Other important factors, where data on decline was lacking but there is other evidence of extreme threat.

Examples of **priority species** in the UK are shown below.

Birds	Terrestrial mammals	Herptiles
■ Common starling. ■ House sparrow. ■ Lesser spotted woodpecker. ■ Scottish crossbill. ■ Skylark.	■ Brown long-eared bat. ■ Dormouse. ■ Otter. ■ Red squirrel. ■ Water vole.	■ Adder. ■ Common lizard. ■ Common toad. ■ Great crested newt. ■ Natterjack toad.
 Lesser spotted woodpecker	 Red squirrel	 Great crested newt

5.7 Protected species and habitats

Several hundred species of birds, wild creatures, insects and plants are protected under the Wildlife and Countryside (Amendment) Act and the Wildlife (Amendment) (Northern Ireland) Order and are listed in various schedules to the Act, for example:

- Schedule 1: Birds
- Schedule 5: Animals
- Schedule 8: Plants.

! Under the Conservation of Habitats and Species Regulations, a person guilty of an offence under the regulations is liable for summary conviction to imprisonment for a term not exceeding six months, or a fine, or both.

The habitat regulations include the following.

- Protection of certain species of animals, Schedule 2 (such as the great crested newt, dormouse, bat, otter and large blue butterfly).
- Protection of species of plants, Schedule 4 (such as the fen orchid and early gentian).
- Designation of Special Areas of Conservation (SACs) and SPAs, which are intended to protect the habitats of threatened species of wildlife.

Local planning authorities will, in consultation with Natural England, Natural Resources Wales, NatureScot or Northern Ireland Environment Agency, consider SPAs to protect birds from the effects of disturbance, shooting, egg collecting or other work.



Under the Conservation of Habitats and Species Regulations, a person guilty of an offence is liable on summary conviction to imprisonment for a term not exceeding six months, or a fine, or both.

5.7.1 Badgers

The Protection of Badgers Act 1992 is provided to protect badgers from deliberate harm, injury or baiting. It is an offence to:

- disturb a badger when it is occupying a sett
- interfere with, damage or destroy a sett
- dig for a badger
- cause a dog to enter a badger sett
- obstruct access to, or entrance to, a sett
- wilfully kill, injure, entrap or ill-treat a badger.

Failure to comply with the legislation can lead to six months' imprisonment and an unlimited fine. A licence must be obtained from the relevant conservation body for any work that may cause disturbance to a badger or involves the damage or destruction of a sett. What constitutes disturbance depends on the nature of the activity proposed. As a rule, a licence is normally required for any work within 30 m of a badger sett. However, this distance may increase for more disruptive activities, such as blasting or pile driving.

Licences are not normally issued during the breeding season, and cannot be issued retrospectively. February is the peak month of the badger mating season, but they can mate any time of the year. It mainly occurs during two periods – one being January to May, and another being July to August. Activities that involve disturbance should therefore be programmed to take place outside of this period.

If a badger sett is discovered after works have started, it is essential to stop work immediately and seek expert advice.



Badgers



Artificial badger sett being built as part of mitigation

The following animals are all protected under the Wildlife and Countryside (Amendment) Act (England, Scotland and Wales) or the Wildlife (Amendment) (Northern Ireland) Order.

5.7.2 Great crested newts

Great crested newts are fully protected by law. It is an offence to:

- capture, kill, disturb or injure great crested newts deliberately
- damage or destroy a breeding or resting place
- take great crested newt eggs
- possess, sell, control or transport live or dead newts, or parts of them
- obstruct access to their resting or sheltering places (deliberately or by not taking enough care).

Each offence could result in an unlimited fine and imprisonment for up to six months.

Great crested newts rely on water bodies for breeding, but otherwise they spend much of their lives on land. They spend winter on land, normally hibernating underground, and emerge soon after the first frost-free days in January or February to begin the migration to breeding ponds (within 500 m). Initial surveys for great crested newts (by an ecological consultant) are required by the local planning authority and Natural England where water bodies are within 500 m of any proposed development. Four surveys must be made to determine presence or absence, with three surveys conducted between mid-April and mid-May.

Ideally, it is best if great crested newts and their habitats are protected before planning permission has been given for development of a site. If a known or suspected great crested newt site is threatened by a development, the local planning authority and the local office of the statutory nature conservation organisation (SNCO) should be informed as far in advance as possible and the appropriate mitigation scheme approved as part of the planning permission.

Any mitigation proposals will need the submission of a licence application to Natural England, Natural Resources Wales, NatureScot or Northern Ireland Environment Agency. It requires a report that sets out survey results, impacts and a mitigation scheme. The application will take up to 40 days to determine.

e.g. Mitigation for great crested newts

Minor impact – on site mitigation.

- Small scale relocation and exclusion.
- Fence erecting and traps set up to exclude newts from works (between March and October). Survey for 30 days minimum.
- Habitat creation (such as creation of refuges and hibernacula – piles of rubble/logs buried beneath the surface of the ground).

Major impact – translocation of newts away from site.

- Fence erecting and traps set up (between March and April). Survey for 60 days minimum.
- Translocation of newts to an area providing equivalent or better habitat (receptor site).
- Habitat creation and restoration prior to translocation (such as areas of coarse grassland, hedgerows and ponds).
- Creation of refuges and hibernacula.



Great crested newt fencing



Great crested newt fencing and mitigation pond

5.7.3 Bats

All bat species, along with their breeding sites and resting places, are fully protected by law. Bats roost in a number of locations; some examples are shown below.

- Disused buildings and structures.
- Under bridges.
- Holes and cracks in trees.
- Roofs and walls of buildings.
- Caves.

Bats hibernate between October and April and breed between May and September. If it is suspected that bats may be present then a bat survey should be carried out by a qualified specialist to establish the size and location of any roost.

It is illegal to:

- deliberately capture, injure or kill bats
- damage or destroy a breeding or roosting place (even if it is unoccupied at the time)
- obstruct access to their roosting or sheltering places
- possess, sell, control or transport live or dead bats, or parts of them
- intentionally or recklessly disturb a bat while it's in a structure or place of shelter or protection.



Man-made bat house to encourage alternative roosting during construction works

Each offence could result in an unlimited fine and imprisonment for up to six months. Personnel holding a bat licence are legally allowed to enter a bat roost or to capture, handle or relocate bats.

5.7.4 Nesting birds

All birds, their nests and eggs are protected by law. It is illegal to:

- intentionally kill, injure or take wild birds
- intentionally take, damage or destroy a wild bird's nest while it's being used or built
- intentionally take or destroy a wild bird's egg
- possess, control or transport live or dead wild birds, or parts of them, or their eggs
- sell wild birds or put them on display for sale
- use prohibited methods to kill or take wild birds.

Some birds, for example barn owls, have additional legal protection. For these bird species, it is also an offence to:

- intentionally, or by not taking enough care, disturbing them while they're nesting, building a nest, in or near a nest that contains their young
- disturb their dependent young.

Each offence could result in an unlimited fine and imprisonment for up to six months.

If any type of bird is nesting in vegetation such as a tree or hedge, then it is legally protected, so long as the nest is still in use. This prohibits certain types of tree pruning or removal until the nesting season is over. Furthermore, removing dense hedges or substantially reducing them is illegal, but annual trimming may be acceptable.

The nesting season is in the spring and summer, generally from the beginning of March until the end of July. However, certain wild bird species such as crossbills may lay their eggs as early as January to coincide with food availability (such as pine seeds), while pigeons and doves nest all year round.

It is essential that construction is carefully planned to avoid any works in areas during the nesting period. If work cannot be avoided during the nesting period, areas must be fenced off to avoid damage or disturbance. There is also a risk of finding birds nesting in dry stone walls and occasionally on pieces of equipment (such as dumpers or cranes) during the construction phase.

Good awareness is important, and site personnel should be made aware of these risks through induction and toolbox talks, along with necessary actions to take.

5.7.5 Trees and hedgerows

Under Part 8 of the Town and Country Planning Act 1990, the Town and Country Planning (Trees) Regulations 1999 and the Planning (Northern Ireland) Order 1991, tree preservation orders (TPOs) can be put in place by the Local Planning Authority to prohibit the cutting down, uprooting, pollarding, topping, lopping, wilful damage or destruction of trees without the local planning authority's consent.

It is illegal to cut down or alter any tree under a TPO. All trees within a designated conservation area are protected.

The Local Authority Planning Department should be contacted well in advance of any work on any trees with a TPO. To work on a tree with a TPO, permission must be sought from the Local Planning Authority by submitting a standard application form. The form is available from the Planning Portal or the authority.

The information on the form must make clear what the proposed work is, and provide adequate information to support the application. The Local Authority has up to two months to make a decision.

Under the Hedgerows Regulations (England and Wales only), a hedgerow removal notice is required from the Local Authority if the hedgerow is older than 30 years and meets at least one of the criteria listed in Part 2 of Schedule 1.

These criteria focus on the length, location and importance of the hedgerow. The removal of a hedgerow longer than 20 m requires planning permission.

During the planning of a project you should make contact with the Local Authority to ensure that any trees or hedgerows identified for removal are discussed and their removal is avoided, where possible.

Translocation of mature or ancient hedgerows should also be considered. Works on trees and hedgerows are confined to certain times of the year due to nesting birds and bats, which are protected species.



Bad practice: tree damage caused by inappropriate storage of construction waste within the protection zone



All trees should be protected in accordance with *Trees in relation to design, demolition and construction (BS 5837:2012)*

The key requirements are as follows.

The roots of a tree help keep it healthy and upright. Roots tend to be found in the top 600 mm of soil, and usually grow out further than the height of the tree.

If roots are damaged, this can cause irreversible harm and lead to the tree's death.

There are three **tree protection zones** surrounding a tree where certain rules apply to help prevent root damage.

The **prohibited zone** is located 1 m away from the trunk. Unless a full consideration with the Local Authority's tree officer has taken place, no excavations can be carried out in this zone. Materials, plant and spoil must not be stored here.

The **precautionary zone** is four times the tree's circumference. If excavations must be undertaken in this zone, mechanical excavation is not allowed. If there are any exposed roots, then precautions must be taken to protect them. Materials, plant and spoil must not be stored here.

The **permitted zone** is any area beyond the precautionary zone. Excavation work can be carried out here, but with caution and limited use of mechanical plant. Any exposed roots must be protected.

BS 5837 recommends additional control measures relating to construction activities, and that the following practices should be followed when construction work is being carried out to protect trees.

Protected zones – a fenced-off area around the tree should be established to eliminate damage during construction, and the rules for each zone must be applied.

Avoid compaction of soil – soil compaction destroys the soil's pore structure, making it hard for trees to absorb water. This can be caused by storage of equipment, materials, machinery and soil, and by using the area as a thoroughfare by people and vehicles, as well as for welfare accommodation.

Considered excavation – when excavating for foundations and utilities, the welfare of root systems must be considered. It is a common misconception that the root system of a tree extends in a narrow, deep band directly below the trunk. In reality, the roots extend a large distance laterally at a shallow depth, at approximately 60 cm.

Shallow foundations therefore impact root systems. The closer the excavation is to a tree, the more likely the chance of damage. Construction therefore should not take place within the protected zone. This also benefits the building as roots can damage foundations. Where foundations are to be located near major roots they should be carefully bridged.

Ground level change – tree root locations should be acknowledged. Increases in ground level could cause compaction, consequently suffocating shallow and fragile roots. Reduction in ground level can cause severance of roots and altered drainage rates, affecting water availability.

Machinery impact – torn branches, damaged bark and general wounds to the trunk can lead to disease and parasites causing decay of the tree.

Contamination of soil – leakage of chemicals from construction materials needs to be avoided through correct and secure storage. The storage of such substances must not be kept within the protection zone and should be downhill from the tree, ideally 10 m from the protection zone to allow for leaching of materials through the soil.

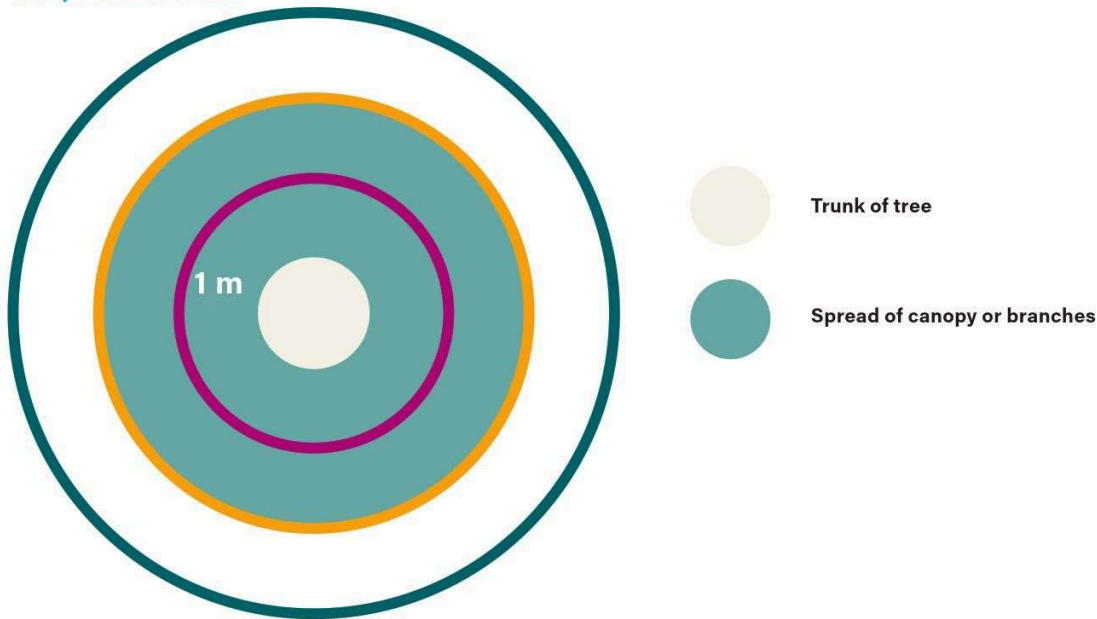
Tree surgery – consent could be required if the tree is protected or is within a conservation area. Any surgery or necessary felling needs to be detailed in the planning application and should be undertaken prior to the commencement of construction activities. It is easier to conduct surgery without unavoidable construction obstacles. Restrictions on surgery and felling must be considered during the bird nesting season.



Good tree protection measures

The National Joint Utilities Group (NJUG) has published *Street works UK guidelines for the planning, installation and maintenance of utility apparatus in proximity to trees*. The document provides details for the establishment of a tree protection zone, and precautions that should be taken for any works within it.

Tree protection zone



Prohibited zone – 1 m from trunk

Excavations of any kind must not be undertaken within this zone, unless full consideration with the Local Authority tree officer is undertaken. Materials, plant and spoil must not be stored within this zone.



Precautionary zone – four times tree circumference

Where excavations must be undertaken within this zone the use of mechanical excavation plant should be prohibited. Precautions should be undertaken to protect any exposed roots. Materials, plant and spoil should not be stored within this zone. Consult with the Local Authority tree officer if in any doubt.



Permitted zone – outside of precautionary zone

Excavation works may be undertaken within this zone. However, caution must be applied and the use of mechanical plant limited. Any exposed roots should be protected.

5.8 Invasive species

An invasive species is an organism that is not indigenous, or native, to a particular area, and can cause significant environmental and economic harm. These can have a damaging impact on British plants, animals and ecosystems by spreading disease, competing for habitat and direct predation (killing of another species for food).

The contract documentation or environmental statement for a project will highlight any invasive species of concern. Identifying invasive species early is key, and if present, advice should be sought from specialists on the best way to treat and dispose of them. Control measures should be included within the project environmental management plan, together with method statements that may need to be agreed with the relevant agencies before commencement of work.

It is possible that construction can impact invasive species, most likely by accidental dispersion. It is an offence under Section 14 of the Wildlife and Countryside (Amendment) Act (England, Scotland or Wales) or under Article 15 of the Wildlife (Northern Ireland) Order to deliberately permit the spread of an invasive species by releasing it into the wild.

One example is the American signal crayfish, which is driving the native white-clawed crayfish to extinction and causing diversity declines in aquatic communities. Commercial fisheries have been affected by egg predation and competition between crayfish and salmon for refuges. Burrowing can cause riverbank erosion and destabilise structures built at the river edges.



Details of invasive species are available on the Government website, including a leaflet on how to manage non-native species.

Section 14 of the Wildlife and Countryside (Amendment) Act and Article 15 of the Wildlife (Amendment) (Northern Ireland) Order makes it an offence to plant or otherwise cause to grow any of the plants listed in Schedule 9. Some common examples include the following.

- Giant hogweed.
- Giant kelp.
- Japanese knotweed.
- Japanese seaweed.
- Himalayan balsam.
- Rhododendron.

- Floating pennywort.
- Parrot's feather.
- Australian swamp stonecrop.

Of these examples, the most common are Giant hogweed, Japanese knotweed and Himalayan balsam.



Refer to Appendix B for details on how to deal with Japanese knotweed.



Giant hogweed



Japanese knotweed (Image by Bridget Plowright)



Himalayan balsam

There are other pieces of legislation that relate to preventing the spread of invasive species in the UK, which can include both native and non-native species (for example, the native species ragwort is one of the species controlled under the Weeds Act 1959).



Practices to be employed for effective management of invasive plants

Before work starts

- Review the environmental statement and conduct a site survey to identify and confirm the location of any non-native invasive species.
- Liaise with landowners and statutory bodies on the appropriate treatment techniques where invasive species have been identified.
- Demarcate and fence off any areas found to contain non-native invasive species.

During construction

- Ensure that workers are made aware of invasive species at site inductions, using maps, and reinforce this via toolbox talks.
- Implement bio-security measures to ensure the work does not contribute to the spread of the non-native invasive species.
- Maintain fenced areas to restrict access and to prevent spread across the site.
- Any use of herbicides near or in watercourses needs to be approved by the environment agencies. Ensure that treatment companies hold appropriate certificates of competence.
- Ensure that any chemical containers or materials contaminated with herbicides are disposed of in accordance with the duty of care at a waste treatment facility in accordance with the Hazardous Waste (England and Wales) Regulations 2005.
- Ensure that soils contaminated with invasive plants or roots are removed to an appropriately authorised landfill site.

Management of invasive species will often generate waste in the form of plant material, soil, contaminated water and sediment, or dead animals. This needs to be dealt with carefully to ensure that it complies with the relevant waste legislation. For example, when Japanese knotweed is dug up, the soil will contain rhizome (underground root-like stems) from which it can regenerate. This is likely to be classified as controlled waste under the Environmental Protection Act 1990, and can only be transferred by an authorised person (such as a registered waste carrier) and taken to an appropriately-authorised landfill site for specialised disposal.



For further information on waste management refer to Chapter E10 Waste and material management.



Pesticides Act 1988 and Control of Pesticides Regulations 1986

In many cases, management of invasive plants will involve the use of herbicides, which are controlled under the Pesticides Act and the Control of Pesticides Regulations. These place strict controls on the supply, storage and use of all pesticides, including herbicides used in the control of invasive species, and make it obligatory for those giving advice on the use of pesticide products (such as salespeople, advisers, managers and those who draft contracts) to hold a certificate of competence recognised by Defra. The only qualifications recognised by Defra are issued by the British Agrochemical Supply Industry Scheme (BASIS). Permission is also required from the relevant environment agency if spraying in or near watercourses.

ECOLOGY – APPENDIX A

Appendix A – Wildlife year planner

Traditionally, environmental issues are not considered at an early stage in project planning. The year planners below show the appropriate times of year to deal with specific issues. These are only a guide, and you must always consult a competent ecologist before carrying out any activity that might disturb these species. You can check this with the Chartered Institute of Ecology and Environment Management (CIEEM). Ecologists should also follow the biodiversity code of practice for planning and development (BS 42020:2013).

Badgers

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1		*	*	*					*	*	*	
2	*				*	*	*	*				*
3	*	*					**	**	**	**	*	*
4		*	*	*								
5							*	*	*	*	*	

- Badger surveys** are best carried out when the vegetation is low and field signs are easier to identify.
- Badger surveys** can continue but become less reliable as the vegetation becomes denser.
- Artificial badger sett construction** between July and October will require a licence to install, and this must be in place for six months before the original sett is closed.
- Badger territorial bait marking surveys.** Territorial marking is at its peak between February and April and the vegetation is low enough to identify latrines. This is the only time of year when bait marking is effective.
- Badger licensing season: Between 1 July and 30 November.** Nature conservancy councils will normally only issue disturbance and exclusion licences outside this time period in cases of proven urgency.

Water voles

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1			*	*	*	*	*	*	*			
2		*	*	*					*	*		

- Water vole surveys** need to be carried out during the summer breeding season (two survey visits are usually required).
- Water vole exclusions.** Excluding and trapping are recommended from 15th February to 15th April, before the breeding season, or 15th September to 31st October, after the end of the breeding season.

Otters

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1	*	*	*	*	*	*	*	*	*	*	*	*
2	*	*	*	*	*	*	*	*	*	*	*	*

- Otter surveys** can be carried out throughout the year, although surveys are easier to carry out during periods of low vegetation.
- Otter mitigation** can be carried out throughout the year but may be restricted if evidence of breeding is identified.

Great crested newts

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1			*	*	*	*						
2			*	*	*	*	*	*	*	*		

- Great crested newts breeding pond surveys** are the only way to effectively establish presence or absence and to quantify populations. These are best carried out between March and June.
- Terrestrial searches** are least effective but can be carried out from March to October, depending upon weather conditions. Surveys should be carried out when night temperatures are above five degrees centigrade and when the ground is moist.

Bats

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1				*	*	*	*	*	*	*		
2					*	*	*	*	*			
3	*	*	*									*
4				*	*	*	*	*	*	*		

- Flight surveys** that involve identification of bats in flight by observation or echolocation can be carried out between April and October, although the optimum time for the surveys is April to September. Surveys of this type can be carried out without a licence as they are non-intrusive.
- Dusk emergence and dawn swarming surveys** can be carried out between May and October although the optimum time for the surveys is between May and September.
- Hibernation roost surveys** can only be carried out between December and March when the bats are hibernating. These surveys are very difficult to carry out as bats hibernate deep in cracks and crevices and are therefore difficult to identify.
- Habitat surveys** can only be effectively carried out between April and October when the bats are active.

Crayfish

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1							*	*	*			
2					*	*						
3	*	*	*								*	*

- Crayfish surveys** are best carried out between July and September, although it is possible to carry out surveys during April.
- Crayfish releasing young surveys** should **not** be carried out during May and June because crayfish could still be carrying their newly hatched young.
- Crayfish reduced activity surveys** should **not** be carried out during the winter months due to reduced levels of activity.

Nesting birds

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1			*	*	*	*	*	*	*			
2	*	*								*	*	*

- Nesting bird season.** No vegetation clearance work should be carried out during the nesting bird season unless immediately preceded by a thorough nesting bird survey.
- Vegetation clearance work** is best carried out at these times of year when birds are not nesting, although work must stop if nests are found.

Reptiles

Task	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1			*	*	*	*	*	*	*			
2				*	*	*	*	*	*	*		
3	*	*								*	*	*

- Reptile surveys** can only be carried out between March and October when reptiles are active, with the peak times April to May and September. During the winter months reptiles are in hibernation, therefore it is not suitable to carry out surveys.
- Reptile capture and release programmes** can only be carried out between April and October when reptiles are active.
- Scrub clearance** work can be carried out from October to February, when the reptiles are hibernating.

Appendix B – Dealing with Japanese knotweed

Japanese knotweed can cause structural damage. It can grow up to 20 mm a day and is strong enough to penetrate foundations. It is an offence under the Wildlife and Countryside (Amendment) Act (England, Scotland and Wales) and the Wildlife (Northern Ireland) Order to encourage the spread of Japanese knotweed. It is often found along railways, riverbanks, roads and derelict sites. Any known areas of Japanese knotweed will be highlighted within the environmental statement for the project. These areas must be cordoned off to prevent inadvertent spread. Extreme care must be taken to ensure that all equipment used on site is free of Japanese knotweed material before leaving the site, to avoid contamination.

If works are required in areas of known Japanese knotweed strands, control methods must be applied; the particular method to be employed will be dependent on site conditions. Where possible, excavation should be avoided and the plant should be treated in its original position.



Japanese knotweed can grow up to 20 mm a day
(Image supplied by Bridget Plowright)



Japanese knotweed in flower



The Japanese knotweed rhizome system (root zone) may extend to, and beyond, a depth of at least 2 m and extend 7 m laterally from a parent plant.

Chemical control

Chemical control usually takes a minimum of three years to totally eradicate Japanese knotweed. Wherever there is a risk of contamination to a watercourse, choice of herbicide is limited to formulations of glyphosate and 2, 4-D Amine that are approved for use in or near water. Use of herbicides in or near water requires formal consultation with the relevant environment agency. Spraying both the top and underside of leaves improves control. Plants respond best when actively growing. The most effective time to apply herbicide is from July to September.

Non-chemical controls

Cutting. Use a simple scythe method of cutting to prevent stem fragmentation. Continue cutting every 2–4 weeks to reduce both above and below-ground biomass (this will only control the knotweed, not eradicate it).

Burning. Controlled burning of stem and crown material may be used as part of the control programme. Such burning must take into account the potential for nuisance or pollution that may occur as a result of the activity. Burning in the open may be undertaken in accordance with a registered D7 exemption from the Local Authority, in accordance with the Environmental Permitting Regulations.

Burial. Soil containing knotweed material and burnt remains of knotweed may be buried on the site of production. On-site burial must be performed to a depth of at least 5 m. Knotweed material should be covered with root barrier membrane consisting of a geotextile layer or a heavy gauge polythene sheet prior to infilling. It is strongly advised to record the burial site location, and to inform any future owners of its position.

Excavation. Only consider excavating as a last resort, unless this is part of an on-site treatment method. When using excavation for off-site disposal, take great care to avoid excess waste and make sure the excavated material does not contaminate surplus soil.

Off-site disposal. Where the option for on-site disposal or treatment is not available, as a last resort in terms of sustainability, material contaminated with Japanese knotweed must be disposed of at an approved disposal facility, having informed the site operator of the presence of viable knotweed within the material.

To ensure compliance with the Wildlife and Countryside (Amendment) Act and to reduce the risk of spreading knotweed, any such on-site burial and any controlled burning must be done in accordance with the new Code of Practice developed by the Invasive Non-Native Specialist Association (INNSA).



For further information on the INNSA Code of Practice visit the INNSA website.



Failure to appropriately dispose of any material containing Japanese knotweed may lead to prosecution under Section 33 and 34 of the Environmental Protection Act and Section 14 of the Wildlife and Countryside (Amendment) Act.

Appendix C – Construction work and its potential adverse effects on wildlife

Construction work and its potential adverse effects on wildlife		
Construction activity	Implication	Examples of effect on wildlife
Site clearance	Removal of trees and shrubs	■ Loss of important species or specimens of tree or shrub that may be protected by a tree preservation order (TPO) ■ Loss of bird nests or bat roosts ■ Loss of habitat for protected species ■ Loss of important invertebrates, including those that may require deadwood habitat (such as stag beetles)
	Removal of ground vegetation	■ Loss of habitat for protected species ■ Loss of rare plants ■ Loss of bird nests ■ Killing or injury of reptiles or amphibians ■ Killing or injury of small mammals ■ Loss of invertebrates and their breeding habitat
	Removal of soil	■ Loss of habitat for protected species ■ Loss of seed bank ■ Loss of water vole burrows ■ Loss of invertebrates and their breeding habitat ■ Destruction of badger setts
	Demolition of buildings and structures	■ Loss of bird nesting or bat roosting areas
	Removal of rubble and other materials	■ Loss of reptile and amphibian habitat
Site set up	Location of site offices and compounds	■ Disturbance of breeding animals
	Storage areas	■ Potential pollution of important watercourses, wetlands or other water bodies, including coastal waters, through spillage or dust
Establishment of haul roads	Rubble or concrete temporary roads constructed	■ Fragmentation of habitats ■ Road kills ■ Destruction of badger setts ■ Contamination of adjoining habitats by dust ■ Noise or light pollution may disturb nesting birds or other animals ■ Change of pH through leaching
Groundworks	Ground investigations Foundations Excavations and piling Temporary earthworks Tunnelling	■ Impacts on surface and groundwater, which may have secondary impacts on important wetlands both on and off site ■ Noise or light pollution may disturb nesting birds or other animals ■ Destruction of badger setts ■ Run-off and erosion, which may damage important habitats ■ Potential to introduce or spread invasive plants (such as Japanese knotweed)
Construction	Concrete pours and other wet trades	■ Silt run-off and wind blown dust create nuisance pollution, including loss of photosynthesis potential for plants, siltation of watercourses and adverse effects on hydro ecology, such as reduced photosynthesis and harm to filter feeders and fish gills, etc. ■ Contamination of wetlands ■ Change of soil pH through run-off

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Statutory nuisance

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Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

STATUTORY NUISANCE

Summary of statutory nuisance legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*				
	EA	LA	NIEA	NRW	SEPA
Acts (primary legislation)					
Clean Air Act 1993	✓	✓		✓	✓
Clean Air (Northern Ireland) Order 1981			✓		
Clean Neighbourhoods and Environment Act 2005	✓		✓	✓	
Control of Pollution Act 1974	✓		✓	✓	✓
Environmental Protection Act 1990	✓		✓	✓	✓
Highways Act 1980		✓			
Pollution Control and Local Government (Northern Ireland) Order 1978			✓		
Pollution Prevention and Control Act 1999					✓
Regulations (secondary legislation)					
Dark Smoke (Permitted Periods) Regulations 1958	✓			✓	✓
Environmental Permitting (England and Wales) Regulations 2016	✓			✓	
Pollution Prevention and Control (Industrial Emissions) Regulations (Northern Ireland) 2013			✓		
Pollution Prevention and Control (Scotland) Amendment Regulations 2017					✓
Guidance					
Considerate Constructors Scheme – Code of considerate practice					
Evaluation and measurement for vibration in buildings (BS 7385-2)					
Net Regs Environmental guidance					
Noise and vibration (BS 5228)					
Process guidance note PG 3/16 for crushing work					
Organisations providing information and guidance on their websites	✓	✓	✓	✓	✓

*Key

EA	Environment Agency
LA	Local Authorities
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
SEPA	Scottish Environment Protection Agency

Overview

Construction sites are inherently noisy. Noise is the largest single source of complaint but consideration should also be given to the effects of certain factors on the local communities neighbouring construction sites, including: lighting; dust; toxic generation; air, water and visual pollution; vibration; bad odours; traffic movements; and operations around vegetation that emit hazardous emissions.

This chapter identifies construction processes that could cause nuisance, provides an overview of the legal framework for their control and provides guidance on mitigating and managing these nuisances on site.

6.1 Introduction



Nuisance is something which is unreasonable that causes substantial interference with people's use and enjoyment of a local environment. It is much more than just an annoyance or being aware of something.

Construction sites (which generate dust, noise, fumes, vibration, heavy traffic movements and artificial light) have the potential to cause nuisance to neighbours. This must be managed effectively to avoid Local Authority or individual intervention, with the potential of delaying or stopping the works.

There are three types of nuisance: statutory nuisance, where a particular nuisance has been recognised in statute law, and public and private nuisance, which are within common law based on precedent (rulings on similar previous cases).

6.2 Important points

Effective planning of site activities should take place to ensure that certain construction activities are limited, so that potential nuisances are minimised and do not cause annoyance to neighbours or impact on wildlife, trees or plants. To avoid causing complaints, sites should adopt the following practices.

- Identify sensitive receptors (people, wildlife, trees, plants and the environment), liaise with Local Authorities on any nuisance that is likely to occur and agree the appropriate mitigation measures.
- Plan all transport activities, both on and off site, to avoid dust, noise, fumes and disruption on local roads.
- Carry out regular dampening down of haul routes and cleaning of the public highway.
- Locate any crushing or cutting equipment away from sensitive areas and dampen down, as appropriate.
- Store dusty materials under cover.
- Consider alternative methods of construction to reduce dust and noise.
- Where noise could be an issue, apply to the Local Authority for a Section 61 prior consent to agree construction methods and noise levels.
- Locate noisy equipment away from sensitive receptors and/or use screening, where necessary.
- Ensure that plant and equipment meet current emission standards and are regularly maintained and kept in good working order.
- Ensure that a monitoring and inspection regime is in place to quickly identify and rectify problems.
- Prepare and communicate the process and responsibilities for dealing with and responding to environmental complaints.

6.3 Legislation

Current legislation for statutory nuisance in England, Scotland and Wales is contained in Part III of the Environmental Protection Act 1990, and is enforced by Local Authorities with controls on various types of noise nuisance regulated under the Control of Pollution Act 1974.

Section 79 of the Environmental Protection Act includes the following statutory nuisances.

- Premises in such a state as to be prejudicial to health.
- Smoke emissions that are prejudicial to health, or a nuisance.
- Fumes or gases emitted from premises so as to be prejudicial to health, or a nuisance.
- Any dust, steam, odour or other waste that is prejudicial to health, or a nuisance.
- Any accumulation or deposit that is prejudicial to health, or a nuisance.
- Artificial light emitted from premises so as to be prejudicial to health, or a nuisance.
- Noise emitted from premises that may be prejudicial to health, or a nuisance.

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- Noise emitted from, or caused by, a vehicle, machinery or equipment on a street or road.
- Any water covering land, or land covered with water, that is in such a state as to be prejudicial to health or a nuisance (Scotland only).

Local Authorities have a duty to inspect their areas from time to time to detect whether a nuisance exists and investigate all complaints of statutory nuisances. Under Section 80 of the Environmental Protection Act, a Local Authority officer can serve an abatement notice, compelling the nuisance to be stopped. Failure to comply with the notice is a criminal offence.

If someone does not comply with an abatement notice, they can be prosecuted and fined:

- a lump sum (the amount is set by the court)
- further fines for each day that they fail to comply (the amount is set by the court).

Councils can also take action to stop or restrict the nuisance by:

- carrying out works and making the person given the notice pay for them (this can include seizure and confiscation of equipment)
- applying to the High Court for an injunction (if a prosecution is not adequate).

Prosecution does not always have to be initiated by the Local Authority. Under Section 82 of the Environmental Protection Act, individuals can also take action through the Magistrates' Court.

All employers should ensure that **best practical means** have been used to prevent, or to counteract the effects of, the nuisance.



Best practical means

Best practical means is defined as taking the following points into account.

- Current technical knowledge.
- The design, construction and maintenance of buildings and enclosures.
- Design, installation, maintenance and periods of operation of plant.
- Financial implications.
- Local conditions.

6.3.1 Airborne dust, emissions and odours

A variety of air pollutants have known or suspected harmful effects on human health and the environment and can create a statutory nuisance. The UK Government has made provisions to ensure that air quality standards for certain pollutants are set centrally through the national air quality strategy.

Local Authorities have a responsibility to carry out and review air quality in their area. Where air quality is found not to meet the national air quality objectives they must declare an air quality management area. The Local Authority must then put together a management plan (called a local air quality action plan) to improve the air quality.

Local Authorities regulate the smaller, less polluting, installations and processes (such as batching plants, crushing work or paint spraying activities). For smaller activities of short duration you may have to apply for a T7 exemption (England and Wales) (mobile crushing and screening) or Paragraph 19 exemption (Scotland and Northern Ireland). For the more polluting processes (such as landfill sites and asphalt plants) air emissions are regulated by the Environment Agency (England), Natural Resources Wales, the Northern Ireland Environment Agency and the Scottish Environment Protection Agency.

6.3.1.1 Controls under the Clean Air Act 1993

The Clean Air Act provides a comprehensive control mechanism for the protection of the environment from smoke, dust, grit and fumes from all fires and furnaces, with the Local Authority as the relevant environmental regulator who can impose limits on dust, emissions and odours generated from a site. Under Section 1 of Part 1 of the Clean Air Act, it is an offence to emit dark smoke from the chimney of any building on any day unless these occur within the permitted periods, as specified in the Dark Smoke (Permitted Periods) Regulations 1958.

6.3.1.2 Controls under Environmental Permitting (England and Wales) Regulations 2016 or Pollution Prevention and Control Act 1999

The provisions of the Clean Air Act do not apply to processes that are controlled under the Environmental Permitting (England and Wales) Regulations, the Pollution Prevention and Control (Industrial Emissions) Regulations (Northern Ireland) 2013 or the Pollution Prevention Control Act in Scotland. Permits are issued for three different types of installation categorised as Part A1, Part A2 and Part B installations. Part A1 installations are regulated by the environment agencies and are for more complex industrial activities. Part A2 and Part B installations are regulated by the Local Authority. From a construction perspective the activities will mainly be classed as Part B installations. These include, but are not limited to, installations such as batching plants, crushing work, waste transfer stations and landfill sites. Businesses which operate these types of premises must have a Part B permit.

The standard conditions for a Part B permit for mobile crushing plant require that no dust must cross the site boundary and the Local Authority must be informed if it does. In addition, visual inspections of dust must be made three times daily.

6.3.1.3 Controls under the Highways Act 1980

Under the Highways Act, Sections 161 and 161A, a person will be guilty of an offence if, without lawful authority, they light a fire on or above a highway, within 50 feet of the centre of the highway, and if they allow mud or other offensive matter to run onto the highway and a user of the highway is injured, interrupted or endangered. Those found responsible for such actions will be liable to a fine, unless reasonable attempts were made to prevent such actions from occurring.

6.4 Dust and emissions

Pollution from dust and emissions can cause problems of nuisance for the local community, businesses and the environment. It can also affect water ecology, such as streams, lakes and ponds; can cause groundwater pollution; and can affect local wildlife and plants.



Managing dust and emissions

Before work starts, an assessment should be made to identify the risk of dust and emissions causing nuisance or environmental damage. The level of risk to local receptors will depend on the following.

- Activities being carried out (for example, demolition, number of vehicles and plant, and types of materials involved in the demolition).
- Duration.
- Size of the site.
- Local weather conditions (for example, wind speed, direction of wind and rainfall).
- Location of receptors to the activities.
- Sensitivity of the receptors to dust and emissions.

If appropriate, liaison with the Local Authority should take place to agree monitoring and control measures where there is a risk that nuisances could occur.

Control measures to mitigate any negative dust impacts could include the following.

- Dampening down of haul routes with water.
- Ensuring public highways are regularly swept.
- Installing wheel-washing equipment at site exits where appropriate.
- Ensuring that bulk materials leaving site are securely covered.
- Installing dust screens (for airborne dust) or silt fencing (for watercourses) to prevent dust spreading.
- Using mist cannons in dust suppression (as shown in the image) and water suppression in cutting and grinding work.
- Ensuring that bulk materials likely to cause dust are covered where appropriate.



Fine water mist being sprayed to reduce airborne dust clouds



For further guidance on the assessment and mitigation of construction dust visit the [Institute of Air Quality Management \(IAQM\) website](https://www.instituteofairqualitymanagement.org/).

6.4.1 Measuring and monitoring dust

All demolition and construction sites should be monitored for air pollution where there is a risk of nuisance or environmental damage. Monitoring and controlling dust levels on construction sites is crucial to the health and wellbeing of all workers, local residents and the local environment. Fine particles of dust can be carried from sites even in light winds. To mitigate the spread of this potentially hazardous substance, a proactive approach is needed.

The need for monitoring will generally depend on existing air quality, air pollution risks from the site, technical practicalities and the financial implications of such monitoring. Deploying best practical means will minimise dust and harmful emissions, which will in turn reduce the need for extensive monitoring.

Although deposited dust is one of the main causes of complaints about air pollution, there are no national UK standards or guidance currently in place. Defra's *Process guidance note PG 3/16* for crushing work does not specify any limits but simply refers to the avoidance of visible emissions crossing the site boundary. This correctly suggests that visual assessment of dust is a main indicator as to whether dust may cause a statutory nuisance. For more accurate assessment, reference is made in unofficial guidelines to a benchmark of 200 mg/m²/day as a value for the threshold for statutory nuisance.



For further guidance from Defra on air pollution and control visit the [Government website](https://www.gov.uk/government).

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6.4.2 Silica dust

Silica is a natural substance found in varying amounts in most rocks, sand and clay. Sandstone contains more than 70% silica, and granite contains 15 to 30%. Silica is also a major ingredient of construction materials such as bricks, tiles, concrete and mortar.

Silica is the biggest risk to construction workers after asbestos. Heavy or prolonged exposure to silica dust is harmful to health, as it can cause serious respiratory diseases such as silicosis and chronic obstructive pulmonary disease.

Many common construction activities generate dust from these materials, including drilling, scabbling, chasing, cutting, grinding, polishing and blasting. The fine dust created is respirable crystalline silica (RCS), and is invisible to the naked eye. This does not remain within the boundaries of the construction site. It will migrate and contaminate the environment on the periphery of the site.



For further information on the health risks associated with silica dust refer to Chapter B10 Dust and fumes (Respiratory hazards).

6.4.3 Diesel exhaust emissions

Most of the air pollution coming from construction sites is related to the burning of diesel fuel. Heavy-duty vehicles, cranes, excavators, generators, and many other types of equipment consume diesel, and discharge emissions into the atmosphere.

The sector's reliance on diesel fuel is due to the absence of grid power at construction sites, specifically in remote geographical locations. Carbon dioxide (CO₂), carbon monoxide (CO), sulphur, nitrogen compounds (NOx) and hydrocarbons are also present in diesel emissions.

The following actions can be taken to minimise the extent of pollution.

- Limit use of road vehicles or other internal combustion engines.
- Ensure that vehicles are switched off when not required.
- Ensure that haul routes are planned with minimum distances.
- Have a planned and preventative maintenance programme or modify existing engines to produce less pollution.
- Use fuels designed to cause less pollution.
- Consider using electrical equipment rather than internal combustion engines.
- Liaise with electricity suppliers early in a project so that equipment can be connected to the grid as early as possible rather than running off generators.
- Use a hybrid solution involving smaller diesel generators and energy storage systems (ESSs) that incorporate Lithium-ion battery technology.
- Use green products such as battery light towers and electric light towers, submersible pumps and mobile compressors.
- Install solar panels to use solar energy to charge tool stations.

Supplementary Planning guidance released by Greater London Authority on *The control of dust and emissions from construction and demolition* contains information outlining the new engine emission standards for non-road mobile machinery (NRMM) in London (under Chapter 7, Cleaner construction machinery for London: a low emission zone for non-road mobile machinery).

These are enhanced standards for monitoring air quality, and are being expanded to other cities in the UK.



For further guidance on processes and procedures related to NRMM visit the NRMM webpage.

6.4.4 Vapours and fumes

Many materials or products (such as sealants or resins) release vapours, fumes or odours. These can damage the environment and be hazardous to the health of workers or other people.

The likelihood of such hazards must be assessed, and adequate control measures designed and implemented that include appropriate monitoring arrangements.

The controls should eliminate the risks, where possible, or otherwise minimise them. Consider selecting less hazardous products to avoid the risks during application and dispose of any used packaging, which will be classed as hazardous waste, appropriately.

6.5 Transport

Vehicle movements and the transport of materials and people represent energy resources being used. It is important that vehicle movements are handled, programmed and managed in an efficient way, as this will lead to greater resource efficiency and reduced costs.

Good logistics, with prompt arrivals and departures and materials being off-loaded in the right locations, will result in multiple benefits. Perhaps the key amongst these will be reduced fuel use and therefore lower carbon emissions, which brings associated cost savings.

Crucially there is also less potential for double handling and wastage of materials.

- There will be less time when noise is being generated. Effective vehicle sequencing will also prevent traffic build-up noise being generated from multiple sources.
- There will be less time when vehicles are on site, with fewer vehicles sitting with idling engines, resulting in lower exhaust emissions. This will reduce the potential adverse impact on air quality.
- Vehicle arrivals and departures that are sequenced in the right way will have less of an impact on local traffic flows and help to prevent local congestion.
- Effective traffic management will result in less dust and mud being produced. This will reduce the amount of road sweeping and dampening down having to be incorporated into the traffic management plan.

The flow of vehicles, materials and resources onto a site needs to be properly sequenced and planned.

This process needs to be undertaken with familiarity with the local environment, such as traffic flows, public road restrictions, wildlife, how the site will be accessed, the size of the site, and adjacent facilities such as schools.

A lack of understanding will result in nuisance to local residents, congestion, delays in material delivery and a variety of negative impacts occurring.

Traffic to and from a site, in relation to materials delivery and people, is one of the main areas of interaction with the local community.

A well thought-out traffic management plan and effective communication of the arrangements will demonstrate consideration and support good public relations with the surrounding community, leading to trust and appreciation in return from local residents.

Depending on where the actual work will take place on streets or roads, registration for a Local Authority lane rental or permit scheme may be needed. Requirements of the permit may push work into off-peak times during the night to reduce the impact on traffic at the busiest times.

This could, however, increase the possibility of creating a nuisance for local residents.



A good traffic management plan will give consideration to the following set of factors:

- Local traffic conditions, peak flows and congestion hot spots.
- Delivery and departure sequencing and times for all site vehicles to minimise traffic movement.
- Off-site delivery and departure routes for all site vehicles.
- Signage and directions on site.
- Hold areas for vehicles waiting to off-load or depart.
- Communication between vehicles and the logistics manager on and off site.
- Wheel-wash locations, dust suppression and mud sweeping.
- Receiving and responding to complaints.
- Consultation processes with emergency services, local residents, schools, public facilities and businesses.
- Properly maintained plant and vehicles to ensure efficiency of operation.
- Consolidation areas and sharing of transport: the use of off-site consolidation areas and the sharing of transport will reduce the number of vehicle movements to and from site. Vehicle sharing can be planned by site staff. Project suppliers may permit the use of their parking areas as car-pooling points.



A good traffic management plan will give consideration to the following factors (continued):

- Demarcated traffic routes and pedestrian exclusion zones.
- Secured areas where heavy duty plant and vehicles are operating.
- Integrating a traffic turning circle, or a one-way road system, to eliminate reversing of delivery vehicles.
- Ensuring that all worker and visitors are aware of the traffic management plan, and are abiding by the instructions.
- Clearly identified parking and lay down areas.
 - On-site parking for staff, contractors and visitors.
 - Delivery vehicle and waste removal parking.
 - Vehicle off-loading and lay down areas.
- On-site traffic management (safe routes, dust and wheel wash).
 - Use wheel washes to prevent mud and any contaminated materials getting onto local roads.
 - Water bowsers can be used for dust suppression.
 - Road sweepers can be used to remove any mud or dust accumulations.
- Promotion of opportunities and arrangements for green travel for site personnel and visitors.

6.5.1 Construction logistics and community safety

Construction logistics and community safety (CLOCS) is a national standard brought about by the direct collaboration between the construction and fleet sectors. The mission of CLOCS is to ensure safe construction vehicle journeys, and to meet the following goals.

- Zero collisions between construction vehicles and local residents.
- Improved air quality and reduced emissions.
- Fewer vehicle journeys.
- Reduced reputational risk.

Construction vehicles continue to be significantly and disproportionately involved in life-changing road collisions with cyclists, pedestrians and the community as a whole. Every construction client and contractor should know their incident data on site, but perhaps do not know how many fatal or serious injury collisions occur on journeys outside of the site's perimeter fence associated with their projects.

In line with Regulation 4 of the Construction (Design and Management) Regulations 2015, clients and principal contractors have a duty to ensure that the construction work they procure is carried out, so far as is reasonably practicable, without risk to the health and safety of any person affected by the project, including the wider community and all vulnerable road users.



For further information visit the **CLOCS website**.

6.6 Noise, including consent requirements

Construction and demolition sites are inherently noisy and are often located in residential areas that are normally quiet. They have the potential to create a statutory nuisance in the form of noise and vibration. Excess vibration results in loud noises, which not only affect the quality of life of local residents, but also impact the local wildlife. Noise pollution can drive wildlife species, such as birds and small mammals, out of the area, seriously compromising the health of the local ecosystem.

It can also cause structural damage to nearby buildings and utilities, and create health and wellbeing risks for everyone working on or visiting the site, as well as the general public. Local Authorities may place restrictions on the person responsible for a construction site to observe specified controls designed to minimise noise and vibration nuisance.

The Control of Pollution Act, Section 60, gives Local Authorities the power to serve notices that specify the following.

- The maximum levels of noise that may be emitted from any particular point.
- Provisions for any change in circumstances.
- The type of plant or machinery that may, or may not, be used.
- The working hours when noise may be made.

BS 5228, which deals with noise and vibration control on construction sites, includes good practice to ensure that a Section 60 notice is avoided. It is in five parts, with Part 1 being a Code of Practice for basic information and procedures for noise and vibration control. The Control of Pollution Act and BS 5228 do not specify any limits for construction noise on the basis that a Local Authority knows its area best and should have the best understanding of suitable noise limits.

6.6.1 Application for Section 61 noise consent (prior consent)

Where it is possible that a noise or vibration nuisance will be created, the person responsible for the site should make an application to the Local Authority for prior consent to start work. Such consent applications should, where possible, be made at the same time as planning applications, or as soon as it is known the nuisance cannot be avoided (such as power-floating at night). The Local Authority has 28 days to approve the application.

Applications should contain particulars of the following points.

- The work to be undertaken.
- Location of the works.
- Working hours.
- Proposed methods, and the plant and machinery to be used.
- Proposed steps for minimising noise and vibration.
- Predicted noise levels.

The Local Authority, in granting consent for the works to begin, may:

- attach any conditions they wish to the consent
- limit or qualify a consent
- limit the duration of consent for the works to be carried out
- specify maximum boundary noise level, permitted hours of work, and plant and equipment that may or may not be used.

In the case of works that overrun for legitimate engineering or health and safety issues, you should advise the Local Authority as soon as is reasonably practicable of the reasons for the overrun, and the likely duration of such works. Where there are minor variations in the works featured in the consent application, and where the works to be rescheduled are of a critical nature, you may apply for a variation. This procedure may also be used for minor additional work that was not included in the original application and does not materially affect the predicted noise levels.

The Local Authority should receive applications for a variation within seven days, where practicable, but at least two working days ahead of the start of the works for which the application is made. Where the operation methods for proposed works have to be changed from the terms agreed in the original consent, you must apply for a dispensation at least 14 days in advance of the proposed operation, submitting the following.

- Details of the operation in question.
- Reasons why the operation cannot be carried out within the terms of the consent.
- Proposed working hours.
- Predicted noise and vibration levels at relevant locations.
- Proposed steps taken to reduce noise and/or vibration to a minimum.



It is far better to apply for a Section 61 consent notice and work with the Local Authority to agree how the work will be carried out, than to be stopped following a complaint and have to comply with a Section 60 notice. Once a Section 61 consent has been given, the contractor is protected from action on noise grounds taken by the Local Authority as long as the conditions of the consent are complied with.

6.6.2 Managing noise on site

Before works commence on site the contract documentation needs to be reviewed to determine whether specific noise limits at various locations from the site boundary have been specified.

Where this is the case, it is likely that monitoring of noise levels will be required and may also be a requirement of a Section 61 noise consent, where this has been applied for.

The main objective of a noise survey is to gain an informed understanding of the main sources of noise, by using a dosimeter to measure noise levels around the site. Noise and vibration may be excessive both in terms of the amount of noise generated, and the duration of the proposed activities on the site. When carrying out a noise survey on site, BS 4142-approved methods should be followed.



Periodic noise survey

STATUTORY NUISANCE

The control of noise on construction sites can be achieved by controlling and reducing the noise at source.

Methods for **controlling and/or reducing noise at source** include the following.

Consider if the loud noise can be removed altogether. If this is not possible, then it must be controlled at source.

Selection of low noise method. Where possible, methods should be employed to reduce the amount of noise generated in the first place (for example, off-site fabrication of concrete panels would avoid the need for scabbling of concrete and the use of vibrating pokers).

Use engineering controls. This could include, for example, using damping to reduce the vibration of machine panels. Use anti-vibration mounts or flexible couplings, avoid metal-on-metal impact by lining chutes with abrasion-resistant rubber, fit silencers to air exhausts, and ensure the better fitting of guards on machinery.

Working hours. Adapt working hours to restrict noisy activity to certain periods of the day. Arrange delivery times to suit the area. Limit the time spent in noisy areas to reduce noise exposure.

Selection of quieter processes or low noise equipment. For some noisy operations (such as piling) there are alternative methods available (for example, drop hammer piling could be replaced by hydraulic jack if ground conditions are suitable). Many power tools are now available that can be operated using electricity or compressed air rather than petrol or diesel engines. Introduce a 'buy-quiet' purchasing policy for machinery and equipment.

Layout and design of the working environment. Design the workflow to ensure that noisy machinery is kept out of areas where workers spend most of their time, and away from quieter areas. Use absorptive materials (such as mineral wool or open cell foam) to reduce the reflected sound. For outside areas, use noise reduction landscaping such as plants and trees. Sound is absorbed by all parts of the plant or tree, such as leaves, branches, twigs and bark.

Provision of acoustic enclosures. Most modern equipment (such as compressors or generators) will come with its own hood or door. These should always be kept closed and in good order. Alter the direct path of the sound through the erection of enclosures around machinery, the use of barriers and screening to block the sound, and positioning noise sources away from workers.

Methods for **screening noise** to create a sound attenuation barrier include the following.

- Site hoarding.
- Purpose-built screens, barriers or walls.
- Placement of material storage areas.
- Bunding.
- Use of existing structures.

The screening should be placed close to either the source or the receptor. Other general measures that can be employed to reduce noise levels are shown below.

- Planning site haul routes to avoid vehicles reversing.
- Planning delivery times and routes to suit local conditions.
- Maintaining haul routes in good order to prevent vehicle noise caused by potholes or uneven surfaces.
- Minimising drop heights of materials into lorries and dumpers.
- Shutting down plant when not required.
- Using only plant conforming with relevant standards and directives on emissions.
- Maintaining plant in good order, including compressor air lines.
- Placing material handling areas away from sensitive receptors.
- Making use of noise-reducing equipment (such as jackets, shrouds, hoods and doors) and ensuring that they remain closed when the equipment is in use.
- Ensuring all viewing openings in site hoardings are glazed with Perspex.
- Hold regular toolbox talks to provide and obtain feedback from workers on the effectiveness of current noise controls.
- Raise awareness of noise via targeted bulletins and good practice guides.
- Regular maintenance and lubrication of plant, equipment and machinery.
- Replace worn, loose or unbalanced machine parts.
- Provision and monitoring of training to relevant workers.
- Continual monitoring of personal protective equipment to make sure it is adequate and being worn and maintained correctly.



Site hoarding that acts as an acoustic barrier

6.7 Vibration

Vibration is caused by the motion of mechanical equipment or sound waves travelling through solid material rather than through air. High levels of vibration can cause damage to buildings, disturb wildlife and disturb neighbours. With regard to statutory nuisance, in the UK, vibration is considered in the same manner as noise.

The most commonly used standard for environmental vibration assessment for disturbance is BS 6472 *Evaluation of human exposure to vibration in buildings [1 Hz to 80 Hz]*. The British Standard suggests that levels of vibration from 0.1 to 0.2 mm/s (at night) and below in residential buildings would have a low probability of adverse comment.

BS 7385: Part 2 *Evaluation and measurement for vibration in buildings* gives guidance on the levels of vibration above which building structures could be damaged. The standard states that there is a major difference between the sensitivity of people in feeling vibration and the onset of levels of vibration that damage the structure. For residential buildings, the standard states that, for cosmetic damage (such as cracking in plaster work) to occur, a peak particle velocity of some 15 mm/s is necessary at a vibration frequency of 4 Hz; this rises to 20 mm/sec at 15 Hz, and thereafter the limit rises to 50 mm/s at 40 Hz and above.

The potential for vibration is dependent on a number of factors, particularly the distance of the receptor from the source, together with the ground conditions and features within the ground (such as sewers). Piling, vibrating rollers, tunnelling and boring are important work that has the potential to cause vibration. Consideration should be given to the methods used to establish the solution with the least vibration risk. These types of activity should be considered carefully when working near to known archaeological features.

Important questions to be considered when planning the works are listed below.

- Can the activity be done using a different technique that results in lower vibration levels (for example, hydraulic pressed sheet piling rather than impact piling or vibrating methods)?
- Because high frequency vibration causes less damage than low frequency vibration, can the plant be operated in a way that generates less low frequency vibration?
- Can the equipment be isolated from the transfer medium (for example, putting generators on timber mats rather than directly on the ground)?

Before starting work, all sensitive structures and buildings should be identified and surveyed. Checks should be made to see if the work is covered under the Party Wall etc. Act 1996. The survey should include photographic and written records of any of the following existing issues.

- Cracks and their width.
- Levels and verticality of tilting walls and bulges in walls.
- Damage, including broken bricks, tiles, pipework or plaster.

Where some vibration cannot be avoided you should adopt a good neighbour policy and inform local residents and the local environmental health officer. Vibration levels should be monitored during the works using competent, trained staff with the appropriate equipment. Following the works, the same survey should be carried out to confirm that no damage has occurred.



Hydraulic pressed sheet piling

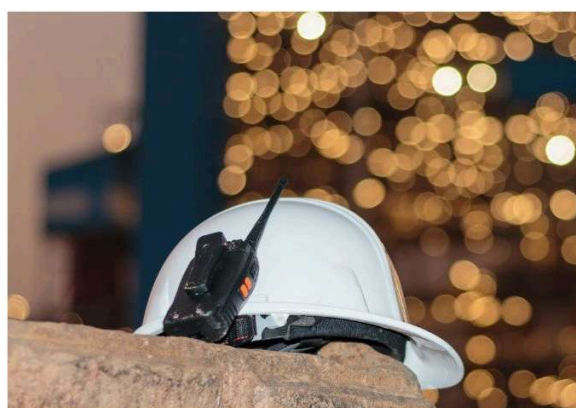
6.8 Light pollution

Light pollution is artificial light that illuminates or intrudes upon areas not intended to be lit. There are a variety of sources of light pollution from construction sites, including lighting towers, offices and access or security lighting.



Light pollution disrupts the natural habitats of wildlife, has negative impacts on human health and compromises the aesthetic beauty of the environment.

In England, Northern Ireland, Scotland and Wales light pollution can be a statutory nuisance and Local Authorities have powers to apply an abatement notice. If this is not complied with the matter may then go to a Magistrates' Court. In Scotland the Public Health etc. (Scotland) Act 2008 introduced measures covering light nuisance that were implemented by amendment to the Environmental Protection Act.



Light pollution can be really disruptive in a number of ways

STATUTORY NUISANCE

Similarly, light nuisance provisions were introduced in Northern Ireland by the Neighbourhoods and Environment Act (Northern Ireland) 2011. Certain areas of the UK are given 'dark skies' status. Often located in national parks or near to observatories, their purpose is generally to promote astronomy. Additional mitigation actions must be taken to minimise light pollution if work is being carried out in these areas.

6.9 Community liaison

A good neighbour has consideration for those around them and takes an active interest in their wellbeing. Construction sites may have many different types of neighbour (such as householders, businesses, sports clubs, pubs and shops) and construction work will affect neighbours in various ways.

Construction projects need to take a proactive approach to community liaison and ensure that the right processes are in place for engaging with the local community. It is important to inform neighbours of what may affect them before it happens so that people can prepare and have a chance to have their say.

There are a number of ways in which this can happen.

- Involving the community (such as local residents, businesses and schools) at the planning and design stage.
- Letter drops to local residents and businesses.
- Visiting owners or occupiers of sensitive homes and businesses.
- Attending local interest group meetings.
- Articles in local publications and newspapers.
- Displaying contact boards with relevant contact details so that interested parties can comment – this is a requirement of the Considerate Constructors Scheme.
- Establishing a complaint system via email or a website.



Local school children get involved in designing their school play areas

It is also important that an effective complaints system is in place that provides a rapid and effective response to issues that have been raised. Being a good neighbour will result in a good relationship with the local community and will avoid complaints and damage to reputation.

The **Considerate Constructors Scheme** (CCS) has supported and driven positive change in the construction industry for the past three decades. The Code of Considerate Practice helps drive forward improved conduct and high standards across construction activities and focuses on three key areas:

1. Respecting the community.
2. Caring for the environment.
3. Valuing the workforce.

There are many organisations that collaborate with the scheme ranging from industry bodies, non-profit/charities and commercial organisations.

The CCS registration has different registration options, and is open to all types of construction activity. All registered site activities are visited periodically by a monitor, whose role is to assess conformance against the scheme's Code of Considerate Practice and give guidance on delivering continual improvements.

The monitor also identifies any measures being taken by a company or its sites that are above and beyond these requirements. Following every monitoring visit, a site or organisation will receive a detailed report capturing findings and highlighting good practice and areas that have been identified for improvement. A Certificate of Performance is also issued, so that achievements against the three areas of the Code of Considerate Practice can be highlighted. Bonus points are available for demonstrating leadership and industry collaboration in regard to the development and implementation of innovations and best practices.



For further information visit the **Considerate Constructors Scheme** website.

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Water management and pollution control



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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

WATER MANAGEMENT AND POLLUTION CONTROL

Summary of water management and pollution control legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*				
	EA	LA	NIEA	NRW	SEPA
Acts (primary legislation)					
Environmental Protection Act 1990	✓		✓	✓	✓
Flood and Water Management Act 2010	Lead Local Flood Authorities				
Water Act 2014	✓		✓	✓	✓
Water and Sewerage Services (Northern Ireland) Order 2006			✓		
Water Environment and Water Services (Scotland) Act 2003					✓
Water Industry Act 1991	✓			✓	✓
Water (Northern Ireland) Order 1999			✓		
Water Resources Act 1991	✓			✓	
Water Resources (Scotland) Act 2013					✓
Regulations (secondary legislation)					
Environmental Permitting (England and Wales) Regulations 2016	✓			✓	
Groundwater (Amendment) Regulations (Northern Ireland) 2016			✓		
Groundwater (England and Wales) Regulations 2009	✓			✓	
Water Abstraction and Impounding (Exemptions) Regulations 2017	✓			✓	
Water Abstraction and Impoundment (Licensing) Regulations (Northern Ireland) 2006			✓		
Water Environment (Controlled Activities) (Scotland) Regulations 2011					✓
Water Environment (Miscellaneous) (Scotland) Regulations 2017					✓
Guidance					
Net Regs Environmental guidance for Scotland and Northern Ireland					
Sustainable drainage systems (SuDS): non-statutory technical standards					
The SuDS manual (C753)					
Organisations providing information and guidance on their websites	✓	✓	✓	✓	✓

*Key

EA	Environment Agency
LA	Local Authorities
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
SEPA	Scottish Environment Protection Agency

Overview

Water is one of our most valuable natural resources and is vital for our survival, our social and economic wellbeing and to maintain precious habitats.

Despite the renewal cycle of water resources, its supply is not endless. Therefore, we have a duty to ensure that it is protected and managed effectively.

This chapter gives a general overview of the legal framework for the protection of natural water resources, including the permit and licence requirements needed for making discharges to foul and surface water drainage systems.

This chapter also provides practical advice for the prevention of pollution to surface and groundwater and implementing an incident response plan should an accidental spillage occur as a result of construction work.

7.1 Introduction

Construction work can cause serious harm to watercourses. Pollution can contaminate drinking water, suffocate fish (by removing essential oxygen from the water) and kill plants, animals and insects living in the water.

A construction site does not need to be next to a watercourse to cause a problem; any pollutants entering a surface water drain can end up in a watercourse miles away.

Pollution by silt can result in the suffocation of fish, destruction of spawning sites and the blocking of drains, which in turn can lead to flooding. Silt pollution can be caused by dewatering, over-pumping, rainwater run-off from uncovered stockpiles of material exposed during earthworks, tunnelling operations, cleaning of ditches and drains, and processes such as wheel washing.

Oil pollution reduces the levels of oxygen in water and can be toxic to aquatic wildlife. It coats plants, animals and birds. Oil pollution is mainly caused through spillages, often from refuelling, but it can be caused by accidental spills, vandalism or the overfilling of equipment.



Oil spreads rapidly: one gallon of oil can completely cover an area of water the size of two football pitches.

Cement and concrete are probably the most common materials used in construction. If either is allowed to enter a watercourse it can have a devastating impact on wildlife.

Cement is highly alkaline and can alter the pH of the water, which can be toxic to aquatic wildlife and contaminate water supplies. Cement and concrete pollution is mainly caused by the cleaning out of equipment and shuttering failure.

Chemical pollution can have a wide range of impacts, including killing fish and plants, destroying habitats and contaminating drinking water. Chemical pollution can be caused by spillages, the leaking of containers or incorrectly-banded areas.

Sewage pollution can be unpleasant, unsightly and smelly, and can decrease the amount of oxygen in the water. Sewage pollution often occurs when drains are blocked, damaged or wrongly connected.

The **Environmental Protection Act 1990** makes reference to controlling the entry of polluting matter and effluents into any place that may ultimately affect a watercourse.

The **Water Resources Act 1991** makes it an offence to contaminate water in a way that may poison or injure fish, spawn, fish food or spawning grounds, or knowingly cause or permit the pollution of controlled waters (such as those listed below).

- Rivers, streams, ditches, ponds, swales, underground streams, canals, lakes and reservoirs.
- Groundwater, wells, aquifers, boreholes or water in underground strata.

It is also an offence to deliberately or accidentally discharge trade effluents into public sewers without the relevant consent. Trade effluents are any liquids produced as part of a trade or industrial activity, excluding domestic sewage.

Trade effluents include the water or slurry from vehicle wheel washes, core drilling, brick/concrete/stone cutting machines, dewatering trenches, pumping out of excavations, concrete washout, pipework cleaning and commissioning, and any similar work.

Mitigation and monitoring processes are essential components to ensure watercourses are protected. Mitigation takes the form of having the necessary preventative measures in place. Monitoring ensures that these measures are working efficiently.

Monitoring must be undertaken on a regular basis to take account of changing site conditions or work methods and inform improvements. In periods of heavy rain, monitoring should be more frequent.



Just half a teaspoon of soil in a bath full of water would be comparable to water quality that could kill fish and smother plants in a watercourse.

7.2 Important points

- Before any work starts on site it is essential to identify all existing site drainage systems and clearly mark them on site plans. Distinguish which are surface water, foul water and combined systems (foul and surface water in the same sewer).
- Seek to install permanent drainage systems as early as possible, as these can then be used to avoid temporary discharges to surface water.
- All drains should be covered or protected to prevent accidental ingress from mud and silt.
- All stockpiled materials should be stored away from drainage systems and watercourses and protected using geotextile silt fencing or cut-off ditches where appropriate.
- The abstraction of water from surface water or piped mains (using a standpipe) will require consent from the Environment Agency (EA) in England, Natural Resources Wales (NRW), the Northern Ireland Environment Agency (NIEA), the Scottish Environment Protection Agency (SEPA), or the relevant water authority respectively.
- All discharges to foul water drainage systems require permission from the local water authority and the consent conditions must be strictly complied with.
- All discharges of effluent to surface water systems will require an environmental permit from the relevant environment agency and the permit conditions must be strictly complied with.
- In England and Wales, temporary (less than three months) discharges from dewatering excavations can be carried out without the need for an environmental permit but strict conditions have to be complied with.
- Where septic tanks are designed to discharge to ground the liquid effluent must pass through a drainage field; you cannot use a soakaway. This will require an environmental permit for groundwater work. Discharges from small domestic septic tanks (that discharge less than 2,000 litres per day) are exempt, providing they meet the EA's general binding rules (England). In Wales the exemption should be registered with NRW and in Scotland SEPA. It is advisable to make contact with the relevant EA to check that your septic tank is not located near a designated sensitive area, and to find out whether you require a permit.
- Before any discharges are made to surface water systems, the water must be unpolluted and free from silt. Silts can be removed through a variety of techniques, as listed below.
 - Settlement tanks.
 - Lagoons.
 - Filtration systems, including the use of gravels, geotextiles or straw bales.
 - Use of flocculants.
- All fuels and chemicals should be stored on impervious material away from drains and watercourses. They should be suitably banded to prevent pollution in the event of leakage or spillage. Refuelling should also be carried out at designated locations with impermeable surfaces away from drains or watercourses.
- All water from vehicle and boot washing facilities should be removed to foul water drainage systems (with the consent of the water authority) or taken away by tanker (waste duty of care must be complied with).
- Concrete and cement washout should be carried out in designated areas. The waste water should not be allowed to enter surface water systems (*further details on the disposal options for concrete washwater and the Environmental Permitting (England and Wales) Regulations 2016 are covered later in this chapter*).
- Monitoring of all permitted discharges should be made on a regular basis to ensure that consent conditions (quality and quantity) are being complied with. Oil and chemical storage facilities should also be inspected.
- Site and public roads should be regularly swept to reduce silt and mud entering surface water drainage systems.
- An incident response plan should be implemented identifying the following.
 - Type and location of drainage systems.
 - Type and location of spill kits.
 - Responsibilities for site personnel.
 - Awareness of environmental issues via training and induction.
 - Arrangements for spill kit replenishment.
 - Arrangements for disposal for contaminated materials from used spill kits.

7.3 Groundwater

Groundwater is the largest available reservoir of fresh water and accumulates underground in gaps in the rocks called aquifers. Groundwater provides a third of our drinking water in England and Wales and also maintains the flow in many of our rivers. In some areas of southern England, groundwater supplies up to 80% of drinking water through the taps.

The Water Resources Act gives the EA or NRW a duty to protect the quality of groundwater and to protect it as a valuable water resource. The Environmental Permitting (England and Wales) Regulations and the Groundwater (England and Wales) Regulations) 2009 implement the requirements of the EU Groundwater Directives. Similar legislation is in place in Scotland and Northern Ireland.

A groundwater protection position statement is available from the EA, offering guidance to stakeholders (such as developers, planners, environmental permit holders and water abstractors).

As the regulator, the EA has established groundwater vulnerability maps and source protection zones (SPZs) for 2,000 groundwater sources (such as wells, boreholes and springs) used for public drinking water supply. There are four levels of SPZs (shown below). The shape and size of a zone depends upon ground conditions, how the groundwater is removed and other environmental factors.



Groundwater sources are one of the largest water sources

Inner zone (SPZ1). This zone includes the 50-day travel time of pollutants to source, with a 50 m default minimum radius.

Outer zone (SPZ2). This zone includes the 400-day travel time of pollutants to source, with a 250 or 500 m minimum radius around the source, depending on the amount of water taken.

Total catchment zone (SPZ3). This is the area around a supply source within which all the groundwater ends up at the abstraction point, or where the water is taken. This could extend some distance from the source point.

Extended zones beneath protective cover. Some zones are extended to include areas where there is protective geology cover, such as clay. This is because activities below the surface, such as deep drilling, could create pathways for pollutants to enter the groundwater. They are indicated on the EA's 'MagicMap' as zones 1c, 2c and 3c.

Zone of special interest (SPZ4). This zone is where local conditions require additional protection.

Groundwater can be polluted by a range of construction materials, including fuels, chemicals, solvents, paints and other liquids. Pollution of groundwater will lead to prosecution. A small quantity of a pollutant has the potential to affect large volumes of groundwater. Remediating large volumes of groundwater is expensive and can take a long time.

Run-off from rainfall has the potential to pick up contaminants as it moves across a construction site. Contaminants may include particulates as well as hydrocarbons and chemicals. Run-off can penetrate permeable surfaces and infiltrate into groundwater.

Every effort must be made to prevent groundwater pollution, including the following.

- Secure materials storage with bunds protecting liquids stores.
- Suitable spill kits and competent spill teams with regular training.
- Storing materials on impermeable surfaces.
- Use of interceptor drains to catch run-off before it reaches permeable surfaces.
- Cover up all drains.
- Keep the road and footpath to the site clean at all times.
- Monitor and improve the management and disposal of site waste.
- Correctly collect and treat any wastewater that is produced.



The unauthorised discharge of chemicals or sewage to groundwater without a permit is an offence and could lead to prosecution. Spills of diesel fuel and oil can penetrate underground plastic water pipes, contaminating the water and giving rise to odours and taints.

WATER MANAGEMENT AND POLLUTION CONTROL

7.4 Abstraction

Water is often taken from natural sources or water mains to reduce dust on haul roads or to reduce dust during cutting operations. Section 24 of the Water Resources Act states that:



... no person shall abstract water from any source of supply or cause or permit any other person to abstract any water, except in pursuance of a licence granted by the authority.

You should therefore not take water from groundwater, watercourses, lakes, streams or water mains without the permission of the relevant authority. The EA, NRW, NIEA and SEPA are responsible for licensing abstractions from groundwater and watercourses. The local water supply company will be responsible for licensing water that is taken from water mains. For guidance on the need for, and process in, obtaining a licence before abstracting any water, the relevant agency should be contacted (i.e. the EA, NIEA or NRW).

In England, Northern Ireland and Wales, if you plan to take more than 20 m³ (20,000 litres) per day, you are likely to need an abstraction licence from the EA. In Scotland, abstractions of less than 10 m³ (10,000 litres) per day do not require an application for authorisation, as they should be carried out in accordance with the relevant binding rules. Water volumes abstracted should be monitored on a daily basis to ensure that the conditions of any abstraction licence are complied with. This will also allow you to check whether there are any leaks in the system.

7.4.1 Dewatering excavations

Construction dewatering is the temporary lowering of groundwater levels by pumping from wells or sumps to provide stable conditions for excavations below the natural groundwater level. The water being removed may require the appropriate consent for discharge into either the foul water system or controlled waters. Dewatering points must be sited to avoid other pollutants or water of a different chemical composition from entering a body of groundwater.

If, as part of the works, you are dewatering or pumping groundwater that has gathered in an excavation, it does not require an abstraction licence if the water is to be disposed of solely to prevent interference with building operations and does not exceed 100 m³ per day, lasts for less than six months and there is no intervening use between the abstraction and discharge. **Where the abstraction is undertaken within 500 m of a conservation area or within 200 m of a spring, well or borehole the limit is 50 m³ per day.**

The dewatering of surface water from excavations does not have an exemption limit but must not cause damage to a protected site or protected species and must be immediately discharged downstream of the building and engineering works. If, however, you intend to use water from a dewatering operation for dust suppression or pressure testing on site, this will be classed as intervening use and you may require an abstraction licence. You must contact the appropriate Environment Agency if your discharge to surface water is going to be more than 10% of the dry weather flow (Q95 low flow) rate of the surface water and dilution is low, as this may increase the risk of flooding.

Geothermal heating and cooling systems that use groundwater to heat or cool buildings can have an impact on water resources. If you are installing an open-loop groundwater or surface water system you may need consent, a permit and a licence from the appropriate Environment Agency. Closed-loop systems will not require permissions.

7.5 Consents

7.5.1 Discharge consents

Where there is a requirement to discharge effluent from any construction activity to drainage systems, watercourses or rivers and streams, an application for consent to discharge must be made to the relevant authority. The issuing authority will depend on where the discharge is made. For example, discharges to foul sewers are usually regulated by the local water company under the provisions of the Water Industry Act 1991. Here, a discharge consent from the appropriate water company is required, except where all discharges are regulated under an environmental permit (England and Wales) or integration pollution prevention and control (IPPC) authorisation (Scotland). Any discharges to surface watercourses, rivers, lakes or ponds would be regulated by the EA, NRW, NIEA or SEPA.

In England and Wales, consents to discharge to surface water systems are regulated under Schedule 21 of the Environmental Permitting (England and Wales) Regulations. The works covered by these regulations includes discharge or entry to surface waters that are controlled (but not to groundwater) of any poisonous, noxious, polluting or waste matter, or trade or sewage effluent. The term *water discharge activities* also includes any work that results in deposits that can be carried away in water (such as cleaning the bottom of a river channel).

As highlighted earlier, temporary discharge from dewatering excavations does not require an application for an environmental permit provided that all of the following apply:

- discharge is temporary, for an overall period of less than three consecutive months
- discharge is made directly to a surface water (such as a river, stream or the sea)
- discharge does not pollute the surface water or adversely affect aquatic life
- discharge location is more than 500 m upstream of a river, a European marine site or a Site of Special Scientific Interest (SSSI), and is not within a site designated for nature conservation (such as National Nature Reserves (NNRs), Local Nature Reserves (LNRs) and local wildlife sites)
- discharge does not cause flooding from the surface water
- discharge does not cause erosion of the banks or bed of the surface water.

Before starting work on site you must:

- plan how to minimise the level of contaminants (such as silt) entering the excavation
- plan how to dispose of water that enters the excavation
- plan not to use machinery in excavations while dewatering is taking place
- minimise water entering the excavation (for example, from rainfall, run-off, groundwater ingress or high water table)
- consider using sustainable drainage construction methods.

Discharges of uncontaminated surface waters are not classed as a water discharge activity. However, you should discuss any proposed discharge of surface water with the regulator before any work takes place.



Make sure contaminated water is contained

Water discharge activities that meet certain conditions are exempt from requiring a permit. Some of these are shown below.

- A small sewage treatment plant discharging 5 m³ or less of effluent per day and subject to meeting other stringent requirements.
- A septic tank discharging 2 m³ or less of effluent per day.
- Vegetation management activities.

Low-risk sewage discharges that meet the relevant conditions can be regulated under an exemption or standard permit. However, all other applications for a discharge permit will be regulated under a bespoke permit.

An application for a **discharge permit** must include, but not be limited to the following.

- The place at which the discharge will take place.
- The nature and composition of the material to be discharged.
- The maximum amount of material that is likely to be discharged in any one day.
- The time period over which the discharge will take place.
- Details of any monitoring and testing arrangements.

In Scotland, discharge consents to surface and groundwaters are regulated through the Water Environment (Controlled Activities) (Scotland) Regulations 2011 (as amended) (CAR).

CAR introduces three levels of authorisations proportionate to the type of risk for the activity.

1. General binding rules.
2. Registrations.
3. Licences.

General binding rules (GBRs), set out in Schedule 3 of CAR, represent the lowest level of control and cover specific low-risk work. Work complying with the rules does not require an application to be made to SEPA, as compliance with a GBR is considered to be authorisation. Since the operator is not required to contact SEPA, there are no associated charges. SEPA has prepared a practical guide to implementing CAR. In addition SEPA is also insisting that a pollution prevention plan must be submitted to them for approval prior to any work commencing.

In Northern Ireland, discharge consents are managed by the Northern Ireland Water Management Unit under the Water (Northern Ireland) Order.

7.5.2 Works consents

Responsibility for regulating work on ordinary watercourses in most areas of England and Wales is controlled by Lead Local Flood Authorities (LLFAs).

These are unitary authorities who lead in managing local flood risks (i.e. risks of flooding from surface water, groundwater and ordinary (smaller) watercourses). They also ensure co-operation between the risk management authorities in their area.

In England and Wales a flood defence consent is required from the LLFA before building a flow control structure (such as a culvert or weir) on an ordinary watercourse. For consent to carry out any works within 10 m of a watercourse, an application must include plans, sections and details including any environmental mitigation measures.

From April 2016 the requirement to have flood risk consent on a statutory main river in England and Wales was transferred to the Environmental Permitting (England and Wales) Regulations 2016. You will require an environmental permit or exemption for flood risk activities from the EA or NRW to carry out any work in, under, over or adjacent to a statutory main river.

If you are currently still working under a flood defence consent, it should automatically be covered as an environmental permit, but it is worth checking with the EA to confirm.

WATER MANAGEMENT AND POLLUTION CONTROL

In Northern Ireland works consent is required from the Rivers Agency, which is part of the Department of Agriculture and Rural Development. In Scotland, as with discharge consents, if you carry out building and engineering work that significantly affects the water environment, this is regulated through CAR and you must comply with one of the following three levels of control:

- comply with certain GBRs that apply to low-risk work
- register your work with SEPA
- get a licence from SEPA.

7.5.3 Septic tanks

The discharge of any sewage from a septic tank to the ground will require an environmental permit for groundwater work unless it is a small domestic discharge, then it may be exempt. In Wales it should be registered under an exemption with NRW. Septic tanks with discharges of less than 2 m³ per day are exempt from environmental permitting as highlighted above.

If any part of the building your septic tank serves is within 30 m of a public sewer, the EA will not allow you to start a new discharge from a septic tank or small sewage treatment plant under the general binding rules.

7.6 Disposal

Drains on site should be clearly identifiable as surface water drains (marked blue), foul water drains (marked red) or combined sewers (marked red 'C'). Surface water drains carry uncontaminated rainwater directly to a stream, river or soakaway. Foul water drains carry foul water directly to a sewage works for treatment before being discharged to a watercourse.

7.6.1 Disposal of contaminated water from dewatering excavations

Water may enter an excavation from either surface water inflows or inflows of groundwater. The inflows may already be polluted or they may pick up pollutants contained within the excavation. The amount of water that is pumped out from an excavation can be reduced by reducing inflows of both surface and groundwater. Edge drains connected to sumps can intercept surface water flows. Cut-off ditches and well dewatering will reduce groundwater inflows.

Any pumping out into a trade effluent system (such as a sewer) will require consent from the water company. An environmental permit from the EA, NRW or NIEA will be required for a discharge to controlled waters (such as rivers, streams or lakes).

However, in England and Wales you do not need an environmental permit for temporary water discharge if the discharge is for less than three months and if you can meet the requirements of the regulator's position statement (or, in Scotland, the general binding rules) for *Temporary water discharges from excavations*.

The regulator's position statement is extensive and highlights issues relating to the water being uncontaminated, measures required to avoid pollution occurring, reducing the amount of water entering excavations and ensuring that the discharge is not near to sensitive protected areas.



Water being pumped to two blue silt traps (top left) allowing silt to settle prior to disposal



For detailed guidance on temporary dewatering from excavations to surface water visit the Government website.

If you need a permit to discharge to a watercourse, it can take up to four months to obtain from the EA or NRW (*for further details of consent requirements refer to 7.5*). Treatment of water before discharge will reduce the potential impact that it has.

Some forms of treatment are shown below.

- Pumping to grassland or other soakaway well away from excavations to avoid recirculation (this option is only suitable for unpolluted water containing only silt).
- Pumping to a settlement tank or lagoon, maximising retention time.
- Using a sump at the base of an excavation, wrapping the end of the pump in aggregates and keeping it off the excavation floor.
- Passing through a filtration system (such as aggregates, geotextile or straw bales).
- Using flocculants (chemicals used to aid solids removal) in conjunction with a settlement tank. Consultation with the regulators should take place first as adding chemicals could make things worse.

Where there is no alternative, water may also be taken and treated off site as waste and will therefore need to be controlled in accordance with the waste duty of care, together with the completion of waste transfer documentation.

7.6.2 Concrete and cement washout

Concrete and cement washout is highly alkaline and can cause severe pollution. Effluent produced from washing out any concrete mixing plant or ready-mix concrete lorries must not be allowed to flow into any drain, watercourse or to ground.

Washout areas must be situated at least 10 m away from watercourses, storm drain inlets, open drainage facilities and trees (roots absorb water), but as close to the works as possible to minimise the risk of spills and drips into the ground.

A lined skip can be used to place waste concrete and effluent, with the water being pumped to a foul sewer or taken away by tanker. Consideration should be given to recycling the concrete washout for reuse in the works.



A clearly designated washout area

7.6.3 Disposal of sewage

The provision of welfare facilities at fixed and transitional construction sites requires that disposal of sewage must be considered. Where possible, disposal may be made by direct connection to a foul sewer. Direct connection to the foul sewer will require consent from the maintaining authority.

On greenfield sites or sites remote from live foul sewers there may be no opportunity to connect to a local foul sewer, in which case a septic tank should be provided that can be regularly pumped out by a liquid waste disposal company. Where sewage waste is taken from site then the waste duty of care must be complied with, and waste transfer documentation completed. Both the location and design of the septic tank are important considerations to avoid raw sewage from entering groundwater or watercourses.

Where effluent from septic tanks is designed to be discharged to ground, an environmental permit will be required. The use of portable toilet facilities should be discouraged wherever possible.

7.6.4 Disposal from vehicle and boot washing



Site boot cleaning facilities



Vehicle wheels being inspected and washed before leaving site

Where wheel-wash facilities are provided on site, the resultant water will be contaminated with silt and possibly oil from vehicle bodies. This must not be discharged to surface water systems, and should be removed to foul sewers with the consent of the local water authority. Modern wheel-wash facilities will allow the wash water to be recycled and recovery of the silts to be separately removed as waste. Where the water is contaminated it may be removed by tanker.

Facilities should also be provided to allow site personnel to clean their boots before leaving site or entering site accommodation. The silty water from these facilities should be dealt with in the same way as the vehicle washing effluent.

7.7 Pollution prevention

A number of measures may be implemented to prevent spillages and reduce the risk of a pollution incident. The EA has a dedicated area on its website to assist in the identification and management of issues of pollution risk in construction and other industry areas. Guidance for pollution prevention provides environmental good practice guidance for the UK, and environmental regulatory guidance directly to Northern Ireland, Scotland and Wales.



For access to EA pollution prevention guidance for businesses visit the Government website.

For guidance on pollution prevention measures in Scotland refer to Sector specific guidance: Construction sites (WAT-SG-75) on the SEPA website.

WATER MANAGEMENT AND POLLUTION CONTROL

7.7.1 Site establishment

When planning the site set-up, the following should be considered.

- Whether the site is in a sensitive area (for example, near to a watercourse or in a site of special scientific interest). If so, restrictions are likely to be placed on the site (such as limited fuel/oil storage).
- Potential drainage on site (such as land drains, foul sewers, surface water drains and soakaways).
- The location of plant and materials away from drains and watercourses, especially fuel storage, top soil storage and waste disposal areas.
- Consider construction of gullies or ditches alongside haul roads and around the perimeter of the working area to collect and channel surface water.
- Whether the environmental regulator (EA, NRW, NIEA or SEPA) or relevant body has given permission to any consents or licences required.
- The placing of stockpiles and spoil heaps, which must be away from drains and watercourses (use geotextile silt fencing or cut-off ditches to avoid silt run-off where appropriate).

When planning or undertaking construction work, consideration needs to be given to the previous use and history of the site and the surrounding areas.

The site on which construction work is to be undertaken may be in the path of ground contamination seepage from an adjacent factory, chemical store, buried waste or other process that may result in pollution.

Records of water pollution may exist, which will give an indication of possible health problems for workers. Obtain all available historical records, as any subsequent pollution may be attributed to the construction company and not to the originators of the pollution.

You should also check if there is any sewage discharge upstream as this may cause issues if you are pumping that water.



Cut-off ditch

7.7.2 Silt management

Good surface water management during construction is essential to prevent sediment pollution, and to minimise the risk of prosecution, which can result in large financial penalties and damaged reputation.

It is essential for the protection of sensitive environmental receptors such as aquatic life in rivers, streams and lakes.

Silt pollution is easily identified by discolouration or cloudy water. Good practices to prevent pollution from happening in the first place include the following.

- Only stripping the minimum amount of land required.
- Use of cut-off ditches and geotechnical silt fences.
- Diverting clean water away from bare ground.
- Not pumping silty water directly into a watercourse.
- Diverting silty water away from drains and watercourses using sand bags, for example.
- Planning for the treatment of silty water when pumping out excavations or managing surface water run-off.
- Having a thorough understanding of site drainage in order to plan how to prevent clean water from entering site as well as how to manage silty water produced on site.
- Retaining vegetation cover, minimise soil stripping and establish new vegetation on bare ground as soon as is realistically possible.
- Identify key areas on site that need protection, such as haul roads, site welfare areas, entrances to drains, at river crossings, in drainage channels, and on slopes.



Silt trap using straw bales and geotextile

Silt can be removed by using silt traps, settlement tanks, ponds or lagoons, by allowing silty water to infiltrate through large areas of grassy ground, geotextiles filters or straw bales (for further information on disposal, refer to 7.6).

7.7.3 Oil and fuel storage

The storage of potentially-polluting materials and the refuelling of mobile plant near watercourses or bodies of water (within 30 m) should be prohibited as far as practicable, and must not be placed where they could be damaged by impact, such as on vehicle turning circles or plant routes. Alternatively, barriers or bollards must be placed around the tank to protect from impact.

Storage should be within an impervious bund with a capacity greater than 110% of the total potential stored contents (for multi-tank bunds, the capacity must be 110% of the capacity of the largest tank or 25% of the total tank capacity, whichever is the greater). The maximum holding capacity should be painted on the side. All level gauges, filling valves and vents and filling nozzles (when not in use) should remain within the bund. All valves should be kept locked when not in use, and available only to authorised and competent persons.



Examples of safe storage for oil

www For further information about oil storage facilities visit the GOV.UK website.

The transportation of fuel across site in drums or other containers should be avoided as far as possible. All mobile plant, including but not limited to cranes, compressors, generators, and so on, should be maintained and operated such that all leaks and spills of oil or fuel are minimised. Oil and fuel storage facilities and storage tanks should be regularly inspected for integrity. Consideration should be given to the actions to take in the event of a spill. Necessary arrangements should be in place and incident response training will be required.

www Further information on oil storage and pollution prevention regulations can be accessed online.

7.7.4 Chemical storage

The floor area used for storing or decanting chemicals must be impermeable. Old or corroded drums will cause more problems than those in good condition. The following measures can be taken to minimise the risk of contamination.

- Start with an up-to-date chemical inventory and Safety Data Sheets, to know what hazards are posed.
- Undertake a control of substances hazardous to health (COSHH) risk assessment and supply it, alongside the Safety Data Sheet, so as to have the correct information for safe storage, use, disposal and what to do in the event of a spillage.
- Purchase chemicals in the appropriate-sized containers to avoid the need for decanting.
- Where decanting is necessary, have safe procedures to avoid any spillage.
- Provide relevant information, instruction, training and supervision to employees.
- Have a proper process for the disposal of all products.
- Provide clear procedures and training for operatives to deal with all types of potential pollution spillages.
- Regularly check for any leaks and arrange immediate repair.
- Have set procedures for the refuelling or replenishing of plant so that any spillage cannot permeate into the ground.
- Install bunding around all storage areas, even temporary fuel stores on construction sites.
- Maintain equipment or storage vessels in good condition.
- Only store or use products that are needed, and in areas with impermeable floors and without drain gullies.
- Maintain good housekeeping procedures and avoid the accumulation of litter or rubbish.
- Ensure chemicals are segregated according to specific hazards.
- Keep containers closed when not in use.
- Shelves in storage cabinets should be level, stable and secured.
- Store chemicals away from direct sunlight, sources of heat, and egress pathways.
- Store hazardous chemicals below eye level.
- Do not store chemicals on the floor, or on window ledges etc.
- Ensure drip trays are in place.
- Label containers clearly.
- Use rated storage cabinets.



An illustration of a chemical storage container

An emergency and incident response plan, appropriate to the size of the site and chemicals being used, should be in place in case of any spillages or pollution alerts (refer to 7.8 for further details).

WATER MANAGEMENT AND POLLUTION CONTROL

7.7.5 Mud

Mud from construction work has the potential to damage the environment. In wet weather it can enter surface watercourses and drains. In dry weather it can dry out and become airborne, in the form of dust, with the potential to be carried some distance and cause nuisance and pollution. Mud can be controlled using road sweepers and wheel washes, and dust can be minimised by dampening down during dry weather.

7.8 Pollution incident response plans (PIRPs)

Emergency planning is a key requirement of an environmental management system, and is necessary under certain legislation. If there is an occurrence of pollution, the polluter pays for the clean up. Pollution clean-up is expensive, especially where immediate action has not been taken to mitigate the effects of the pollution immediately after it occurs. The clean-up costs of contamination could far outweigh those imposed by the courts, as a lot of these costs are hidden, such as:

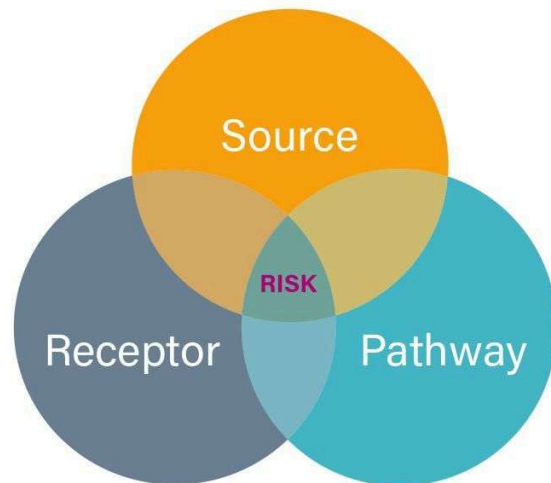
- Replacement or repair to damaged, ruined plant and materials.
- Prosecution costs, e.g. fines and legal costs.
- Costs of civil claims for losses suffered by third parties.
- Loss of reputation, resulting in loss of repeat business.

Prompt action is needed to protect people and the environmental receptors. Incident readiness and response begins with considering what emergency scenarios and incidents (source) may occur on a construction site, on what pathways the pollution can travel and what in the surrounding environment may be impacted by them (receptor).

Types of incident include fires, spills and leaks. For each type of incident several different parts of the environment may be affected. For example, a fire will release products of combustion into the atmosphere, not only damaging the premises but potentially affecting neighbouring properties, the local community and the environment. Water used for fire-fighting may enter surface watercourses and sewers and permeate the ground.

An effective pollution incident response plan should include the following.

- Emergency contact numbers.
- Training of site personnel in the emergency response plans, and their specific roles and responsibilities.
- Practicing the incident response plans, so that staff have a clear understanding and expectation of what is required of them.
- Systems in place to be occasionally practiced to evaluate effectiveness, so that areas for improvement and necessary corrective actions can be identified.
- Easy access to site plans, which show the locations of site drainage, foul, surface water and natural run-off, as well as where materials are stored, the locations of permitted discharge points, bunded storage and secondary containment areas, and equipment emergency shut-down control panels.
- The use of containment equipment (such as drip trays, bunds and booms) to capture and avoid spread of the spillage.
- The location of spill kits and any other emergency equipment around the site, including designated refuelling areas.
- Arrangements and responsibilities for spill kit replenishment after an incident.
- Arrangements for the safe disposal of used spill kit equipment and contaminated materials.
- A relatively safe location from where to control the emergency situation with good sources of communication.
- Specific training for plant and equipment operatives, so that they recognise potentially hazardous situations and can take the necessary actions to mitigate the risks from the environmental hazard.
- For the incident response plans to operate smoothly and efficiently, it is essential that processes and responsibilities are laid down and understood by both internal (relevant company personnel) and external (fire, police, ambulance, HSE, local companies, local or national enforcement bodies) in line with an established environmental reporting procedure.
- Ensuring that a suitable and sufficient record of the incident is recorded and identified actions taken.
- Ensuring that the correct equipment is in place to deal with spills, such as absorbent materials (loose to deal with small spillages, in bags of various sizes to contain spills); flotation tubes to deploy into water sources to contain oil and fuel spillages; sand bags or similar to block drains and divert flows of potentially hazardous substances to containment areas; and temporary portable bunds to stand leaking containers in.
- Access to an up-to-date substance inventory, which should include any innocuous substances that could be environmentally damaging if they escaped, such as tanker milk spilled into a watercourse that will destroy the ecosystem.
- Guidelines on handling the press and media. This should be done by a competent person who has been thoroughly briefed and has knowledge of the organisation and environmental issues. This will ensure that reasoned and accurate responses are given to press and media questions.



A review of the incident response plans after an incident is vital, and processes need to be in place to ensure that this is carried out. Shortfalls in the current arrangements can be identified, and measures put in place to prevent or minimise the consequences of any foreseeable future incidents. This would form part of a management review within the Plan-Do-Check-Act cycle.

It is a requirement of the ISO 14001 standard to strive for continual improvement.

For each type of incident a plan must be put in place setting out the response needed. The basic process of **stop – contain – notify – clean up** must be followed. In the event of a spillage it is essential that a competent spill response team is deployed with the right spill clean-up materials. The team has to respond quickly for the spill to be controlled before it causes any environmental damage.

7.8.1 Booms

If work is to be carried out adjacent to a watercourse or river then arrangements should be put in place to ensure that a boom, which prevents the surface spread of oils and chemicals, can be deployed quickly to contain the spillage. The boom should be long enough to span the full width of the river.

If the work is carried out on navigable waterways then the appropriate authority should be notified before the boom is installed.

7.8.2 Drip trays

Where drip trays are used they should be used in conjunction with absorbent mats so that the contamination can be easily removed. Drip trays should not be left to fill with rainwater, as the resulting contaminated water will be more of a problem to dispose of and could cause contamination of the ground, drains and watercourses if left to overflow. Plant nappies are also available and allow water to pass through while absorbing any oils.

7.8.3 Spill kits

The type and quantity of spill kits deployed around a site will be dictated by the nature of the works and whether they are carried out on land or water. The type of spill kit will also depend on the type and quantity of oils and chemicals that are being used on the site. Typical spill kit equipment could include, but is not limited to, the items below.

- Absorbent booms.
- Absorbent granules.
- Drain covers.
- Heavy-duty plastic bags.
- Absorbent pads.
- Shovel.
- Gloves.



Spill kit deployed

There are multiple types of spill kits used in the construction work environment.

Oil spill kits are used for the clean-up of grease and oil-based liquids. They absorb oil and reject water.

Chemical spill kits are used for common hazardous liquid spills, as they soak up and hold chemicals. They can be used for outside and indoor spills, leaks and drips of corrosive chemicals and hazardous liquids.

General purpose spill kits are suitable for smaller and non-aggressive liquid spills. These are commonly used to absorb water, solvents, coolants, paint, blood, fluids, degreasers and less-damaging chemicals.

Anti-static spill kits are used for the clean up of highly flammable or explosive liquids.

The correct spill kit must be chosen to reflect the potential source of contamination, and this will be determined from the environmental risk assessment, both in terms of type and how many kits will be needed.

Spill kits should be clearly marked and located at signposted locations around the site. Arrangements for the replenishment of the contents should be kept on the inside lid of the spill kit.

Equipment contaminated with oils and chemicals is likely to be hazardous waste and should be stored separately and disposed of in accordance with the duty of care and Hazardous Waste Regulations 2005.

Spill response training and simulated tests should be carried out on a regular basis to evaluate the effectiveness of incident response plans. According to the success of the response, improvements may be needed to the system.

After any incident, management must conduct a post-incident investigation to examine and determine what improvements are needed to prevent the incident from happening again.

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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

RESOURCE EFFICIENCY

Summary of resource efficiency legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*				
	EA	LA	NIEA	NRW	SEPA
Acts (primary legislation)					
Climate Change Act 2008	✓		✓	✓	
Climate Change (Scotland) Act 2009					✓
Environmental Protection Act 1990	✓		✓	✓	✓
Environment Act 2021	✓				
Regulations (secondary legislation)					
EU Timber Regulation	✓		✓	✓	✓
Guidance					
ISO 5001 Energy management					
ISO 14001 Environmental management systems					
ISO 20400 Sustainable procurement guidance					
Organisations with information and guidance on their websites	✓	✓	✓	✓	✓

*Key

EA	Environment Agency
LA	Local Authorities
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
SEPA	Scottish Environment Protection Agency



Refer to Chapters E03, E07 and E10 for specific legislation on energy, water and waste topics, which link into resource efficiency.

Overview

The construction sector is the largest consumer of materials in the UK, and the largest producer of waste. More efficient use of materials would make a major contribution to reducing costs and the environmental impacts of construction, including carbon emissions, landfill and the depletion of natural resources.

This chapter provides a brief introduction to the environmental and economic benefits of improved material efficiency and waste reduction. It provides guidance on the responsible sourcing of materials, chain of custody for timber and life cycle analysis, and highlights the benefits of reusing and recycling materials.

8.1 Introduction

Construction uses vast amounts of natural resources and raw materials. Historically, it has been an inherently inefficient process, arising from the bespoke nature of on-site construction. This not only wastes a lot of money, it also produces high levels of waste materials and causes excess material extraction to replace those that have been lost through inefficient use. Resource efficiency ensures that fewer materials are used (reducing waste outputs) to build or renovate buildings and infrastructure, while still achieving the same – or better – overall functionality. To be resource efficient, the industry needs to use the planet's limited resources in a sustainable way, and at the same time minimise impacts on the environment.

The Design for Manufacture and Assembly (DfMA) engineering methodology is an example of resource efficiency that enables a building to be designed and constructed much more quickly, efficiently and sustainably. It increases construction efficiency by using a manufacturing environment to create elements of a building off site, rather than on site. Standard designs and specifications are used to ensure that design and construction quality are not compromised. DfMA has the potential to create buildings that can be easily disassembled and re-assembled at other locations. This adds to the life cycle possibilities of a building and increases recycling, with the potential to achieve zero waste. It provides both short and long-term benefits such as improved health and safety, increased productivity and quality control, higher resource utilisation, shorter time to market, reduced product development, labour costs and wastage, and greater product reliability.

8.2 Important points

Clients, designers, manufacturers, suppliers, contractors and installers have an important role in ensuring that the materials and products that form a construction project are both sourced and used with sustainability in mind. This is a pre-requisite of meeting certification to standards such as the Building Research Establishment Environmental Assessment Method (BREEAM).



For further details on sustainable construction standards refer to Chapter E01 Sustainable construction and the environment.



Practical ways for improving resource efficiency and reducing waste

- **Design.**
 - Specifying materials from sustainable sources; recycled and locally sourced.
 - Designing the project to suit standard product sizes and to avoid site cutting.
 - Designing for off site, pre-assembly and ensuring that products are responsibly sourced.
 - Designing to allow a cut/fill balance and by utilising surplus materials in site features (such as landscaping).
 - Looking for opportunities to source excess materials from other projects (for example, utilising the Contaminated Land: Applications in Real Environments (CL:AIRE) register of materials and services).
 - Specifying non-hazardous, low-impact materials with low embodied carbon.
 - Designing to allow for disassembly so components can be recovered and used elsewhere.
- **Procurement.**
 - Buying construction products from sustainable sources from suppliers with a good environmental track record.
 - Requiring sub-contractors to have a waste management policy and source sustainable products.
 - Not over-ordering materials.
 - Reducing the amount of packaging.
 - Ordering materials at the size required, to avoid off-cuts.
- **Construction.**
 - Avoiding over-excavation.
 - Storing materials to avoid damage, theft, contamination and double handling.
 - Segregating surplus materials for reuse elsewhere.
 - Crushing existing demolition waste for reuse in the works to avoid the need for virgin materials.

RESOURCE EFFICIENCY

8.2.1 Life cycle assessment (cradle to cradle)

Life cycle assessment (LCA) is a process of evaluating the effects that a product has on the environment over the entire period of its lifetime, thereby increasing resource-use efficiency and decreasing liabilities. A good LCA will identify where in the life of a product the main impacts occur and identify what can be done to reduce or mitigate these impacts. In construction LCA is applied to both the building itself and also to the products that go into that building.

The stages in the life of a building are material extraction, processing into a product, combination of products into the building, use of the building, end of life deconstruction and material recycling. Efficient recycling of materials at end of life has now given rise to the term *cradle to cradle*, while *circular economy* (CE) has gained traction, with many construction companies using the notion to operate in a way that minimises waste.

The CE system is based on the reuse and regeneration of materials or products, as a means of continuing production in a sustainable or environmentally-friendly manner. Unlike a linear economy of make-use-dispose, the aims of CE encompass all the LCA and cradle to cradle criteria with the intention of keeping materials, components and products at their highest utility and value. CE has beneficial impacts on the three pillars of sustainability: **environment**, **economy** and **society**.



For further information on the circular economy visit the Green Construction Board's online resources contained on the Construction Leadership Council's website.

The EMS Standard ISO 14001:2015 requires organisations to adopt a life cycle perspective (Clauses 6.1.2 and 8.1 refer). Annex A to the standard also provides guidance on this approach. ISO 14001:2015 has also introduced the idea of looking for opportunities to reduce environmental impact. Improving resource efficiency is one way of demonstrating this.



For further details refer to Chapter E02 Site environment management systems.

Life cycle costing is a narrower assessment of the overall economic impacts of an asset, whereas the use of environmental costs in a whole-life analysis allows a true comparison between options, particularly where they are quoted as 'good' for the environment. For a major project (such as the construction of a nuclear power station) it is possible to calculate the environmental impact of making the concrete containment, the water required for refining the copper for the power plants and all the other components. Only by undertaking such an analysis is it possible to determine whether one solution carries a lower or higher environmental cost than another.

PAS 2050, *Specification for the assessment of the life cycle greenhouse gas emissions of goods and services*, produced by the British Standards Institute, is a publicly available specification that provides a consistent method for assessing the life cycle greenhouse gas emissions of goods and services.

8.3 Sustainable and responsible sourcing of materials

With the increasing focus on sustainable development, many construction companies are recognising the need to prove that their buildings are built with sustainability in mind. One element of this is in the responsible sourcing of products used in their construction, and the onus of proof is increasingly being passed back to the manufacturers of those construction products.

BREEAM awards credits based on the environmental impact of materials and for materials responsibly sourced. The aim is to encourage the use of materials with lower environmental impacts over their life cycle and to recognise and encourage the specification of responsibly sourced materials for basic building and finishing components.

The Building Research Establishment (BRE) standard BES 6001 has been published to enable construction product manufacturers to ensure and then prove that their products have been made with constituent materials that have been responsibly sourced. The standard describes a framework for the organisational governance, supply chain management and environmental and social aspects that must be addressed in order to ensure the responsible sourcing of construction products.

Independent, third-party assessment and certification against the requirements of BES 6001 then give the organisation the ability to prove that an effective system for ensuring responsible sourcing exists and adds credibility to any claims made. BES 6001 is also aligned to BREEAM so that products from manufacturers certified against this standard are able to score points against the responsible sourcing credits.

The BRE *Green guide to specification* provides guidance on how to make the best environmental choices when selecting construction materials and components. The guide presents information on the environmental impacts of building elements and specifications by ranking them on an A+ to E rating scale, where A+ represents the best environmental performance. These environmental rankings are based on LCAs using an environmental profile methodology. They are generic rankings that illustrate a range of typical materials.

BS 8902 provides a framework and gives requirements for the management, development, content and operation of sector certification schemes for the responsible sourcing and supply of construction products.

For manufacturers and suppliers, having certification to the environmental management systems (EMS) standard (ISO 14001) provides some evidence that the company is measuring, and improving, the environmental impacts of their products. Relevant products assessed under the responsible sourcing elements of BREEAM are able to score points against credits where the certified EMS covers the main processes or main supply chain processes.

For example, the responsible sourcing of bricks would need to demonstrate that the certified EMS covers manufacture (main process) and clay extraction (main supply chain process) to obtain maximum points.

8.3.1 Water efficiency

Water is a precious global resource, critical for life in all its forms. As the world's population increases, there will be an ever-rising demand on what is a finite resource. Future climate change and resulting changing patterns of rainfall will make water supply increasingly challenging through the ageing water supply infrastructure.

Water is an expensive item with costs on both the supply side and the waste water treatment side. Given these pressures water will become a more expensive resource. What this means within construction is that water efficiency will become a higher priority at all stages in a building's life cycle. Materials with high embodied water content (such as high water usage during manufacture or use) will inevitably increase costs to take account of this priority. The increasing importance of embodied water is recognised in the Building Research Establishment Environmental Assessment Method for Infrastructure (BREEAM Infrastructure).

8.3.2 Water scarcity

Water scarcity is both a natural and a human-made problem. There is enough fresh water on the planet for six billion people but it is distributed unevenly and too much of it is wasted, polluted and unsustainably managed. Water scarcity is among the main problems to be faced by many societies in the world. Water use has been growing at more than twice the rate of population increase in the last century, and, although there is no global water scarcity as such, an increasing number of regions are chronically short of water.



Water scarcity is defined as the point at which the total impact of all users affects the supply or quality of water under existing arrangements to the extent that the demand by all sectors, including the environment, cannot be fully met.

Water scarcity can occur at any level of supply or demand and may be caused by human behaviour or the consequence of altered supply patterns (for example, from climate change). Scientists typically measure scarcity by looking at the total amount of water available per person. An area is experiencing water stress when annual water supplies drop below 1,700 m³ per person. When annual water supplies drop below 1,000 m³ per person, the population faces water scarcity, and below 500 m³ is absolute scarcity.

Water scarcity already affects every continent. Four billion people – almost two thirds of the world's population – experience severe water scarcity for at least one month each year. Over two billion people live in countries where water supply is inadequate. Half of the world's population could be living in areas facing water scarcity by as early as 2025, and some 700 million people could be displaced by intense water scarcity by 2030. By 2040, roughly one in four children worldwide will be living in areas of extremely high water stress.

The UK's population is forecast to rise to 75 million by 2050. Most of this population growth is expected to be in geographical locations already experiencing water stress; mainly London and the South East of England. The Environment Agency has warned that within 25 years, the UK will not have sufficient water to meet demand due to climate change (*for further information on water efficiency and minimisation refer to 8.3.1*).

8.3.3 Using and conserving water during site works

On larger construction projects, where high volumes of water are being used, the first action is to establish an approach to measuring and monitoring water usage so that it can be managed. This may involve the use of water meters at appropriate locations and the use of water balances to account for water usage. Water reduction targets can be set based on known volumes of water usage and progress monitored. Collecting data on water consumption during the construction process will provide the following benefits.

- Understanding and managing costs.
- Reducing environmental impact of overuse.
- Benchmarking and improving performance.
- Obtaining credits under BREEAM.
- Demonstrating continual improvement in accordance with ISO 14001/EMAS.
- Demonstrating good practice and meeting customer expectations.

Significant savings can be made by using rainwater harvesting systems to collect rainwater from roofs and other flat surfaces, or other impervious surfaces, so that it can be stored for later use. Early installation of suitable run-off collection systems would need to be investigated at the design stage and payback times calculated for the expected volume of water use. Harvested water can be used for dust suppression, avoiding the need to draw water from the mains or abstraction.



Collecting rainwater for reuse

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Water usage should be monitored and any obvious leaks and running hoses identified and dealt with. The use of triggers on hoses will prevent hoses from running whilst unattended.

Vehicle wheel-wash equipment is now available with water recycling and recirculation systems fitted. These will reduce the volume of water used and have the potential to save money. These systems work by providing a solids settling area combined with the use of flocculants to further precipitate solids out. The solids collected can be periodically removed.

Site welfare facilities can be fitted with waterless urinals, push taps and rainwater harvesting for toilet flushes.



A roof area of 400 square metres will collect over 12,000 litres of water after 25 mm of rainfall.



Waterless urinals

8.3.4 Sustainable drainage system (SuDS)

Sustainable drainage systems (SuDS) offer a process of managing rainfall that minimises the negative environmental impacts on the quantity and quality of run-off water, whilst maximising the benefits of amenity and biodiversity for people and the environment. The philosophy of implementing SuDS on a proposed development has four key pillars:

1. Water quality.
2. Water quantity.
3. Amenity.
4. Biodiversity.

SuDS can be used to manage surface water run-off from large areas (such as part of a housing estate, major roads or business parks). They provide a natural approach to managing drainage in and around developments. SuDS work by slowing down and holding back the run-off from a site, allowing natural processes to break down pollution.

They deal with run-off close to the source rather than transporting it elsewhere. They are designed to attenuate (storing and slowly releasing) surface water from developments in a manner that will provide a more sustainable approach than the previous, conventional practice of routing run-off through a pipe into a watercourse. They are also a tool for preventing flooding. (Early engagement with the lead local flood authority is advisable at the planning stage.)

Facilities for SuDS include the following.

- Permeable surfaces.
- Filter strips.
- Filter and infiltration trenches.
- Swales.
- Detention basins.
- Underground storage.
- Wetlands.
- Ponds.

Other facilities exist (such as hydraulic controls or silt traps).

In **England** the National Planning Policy Framework (NPF) is in place, and is currently being updated in order to develop policy to support levelling up, as well as how national planning policy is currently accessed by users. Currently, under the NPF all major developments and housing projects of 10 houses or more should incorporate SuDS, unless there is clear evidence that it would be inappropriate.

In **Scotland, Wales and Northern Ireland** the rules are considerably different. In Scotland, SuDS are a legal requirement for all developments except single dwellings that drain to the water environment, unless they discharge to coastal waters.

The Scottish Environment Protection Agency (SEPA) is the statutory agency responsible for protecting the water environment, and therefore requires the use of effective, appropriate SuDS features in new developments.

In Wales, all new developments of more than one dwelling house, or where the construction area is 100 square metres or more, will require SuDS for surface water. In Northern Ireland, there is currently only a requirement in the planning process to demonstrate that SuDS have been considered.

Whilst the benefits of SuDS are clear, conditions will be required by the planning authority to ensure that plans are in place and responsibilities allocated for the ongoing maintenance of SuDS for the lifetime of the development. The plans should cover general maintenance of litter clearing, vegetation control and more long-term repairs and/or dredging.

8.4 Timber and chain of custody

Chain of custody (COC) is a process that provides assurance about where timber has been sourced from. This is done by certifying timber from the forest to the final point of purchase and requires an effective audit process.

The process tracks timber through each stage of the supply chain, from forest and logging, through sawmill, factory and distributor, to timber merchant and contractor. This provides transparency and traceability to guarantee compliance with demands for ethically sourced timber products.

Construction uses large amounts of timber, so it is important that timber is sourced from forests that are managed in a sustainable and ethical way. Illegal logging results in soil erosion, loss of biodiversity and uncontrolled development. It is therefore important to know where timber has been sourced from.

The Timber and Timber Products Placing on the Market Regulations 2013 (UKTR) and Forest Law Enforcement Governance and Trade Regulations 2012 (FLEGT) give operators and traders an important role in ensuring that timber is sourced and supplied from legal and verifiable sources. This also applies to virgin wood products used for renewable heat installations (such as biomass boilers).

To combat illegal logging, which has serious environmental, economic and social consequences, timber imported from countries that have implemented a voluntary partnership agreement (VPAs) with the UK must be accompanied by a FLEGT licence. Under this agreement, the timber producing country agrees to control and license its timber exports to ensure that the timber products come from a legal source. These agreements are underpinned by strong systems for ensuring timber legality.

Contractors and suppliers working for the UK Government have to comply with the timber procurement policy (TPP). The policy sets rules that all products derived from timber and wood must be from independently verifiable, legal and sustainable sources, or FLEGT-licensed timber or equivalent sources.

The crucial point that clients and contractors need to know is that any timber used on site for construction or timber hoardings must be legally sourced. The most easily identifiable way of doing this is the **Forest Stewardship Council (FSC)** label, as this indicates that there is a traceable trail of where the timber has come from.

The UK Government has approved the FSC and the **Programme for the Endorsement of Forest Certification (PEFC)**, the two largest international forest certification programmes, as timber certification schemes that meet the central point of expertise on timber (CPET) requirements.

The FSC and their global certification scheme allows consumers to identify, purchase and use timber and forest products produced from well-managed forests. FSC UK is a national office of the FSC, which is an international non-governmental organisation dedicated to promoting responsible management of the world's forest.

In the UK, FSC forest management certification is based on UKWAS, a national standard which is consistent with FSC principles and criteria whilst at the same time reflecting local ecological, social and economic circumstances.

PEFC UK is part of the global network for the assessment and mutual recognition of national forest certification schemes developed by the PEFC Council. PEFC UK's certification scheme for sustainable forest management incorporates the UK Woodland Assurance Standard as the certification standard to be used for assessing whether a woodland is managed sustainably.

For further information on the FSC and PEFC visit the respective websites.

The BRE put responsible sourcing of timber at the top of its agenda in the development of BREEAM, making it a mandatory element within any BREEAM assessment. It also recognises FSC certification as an indicator of responsibly-sourced materials, and has adopted CPET requirements for the purpose of demonstrating responsible sourcing of timber. This should be high on the agenda for any project team involved in a BREEAM assessed project. Some large construction projects have adopted a chain of custody certification scheme themselves as good practice.



Around 10 million hectares of forest are lost each year, an area the size of Portugal



Timber and chain of custody certificate number



PEFC UK forms part of a global network of forest certification schemes

8.5 Reuse and recycling of materials

Maximising the reuse of materials on site can significantly reduce the amount of waste generated. For example, careful cut and fill analysis can ensure that soils excavated from cuttings can be used as fill material elsewhere (such as within embankments) so that no waste is sent to landfill and the need to procure fill is removed.

Not only can demolition materials be processed for aggregates and fill materials, designers should aim to use other materials (such as reclaimed bricks, timber and steel sections).

Other materials that may incorporate recycled content include plastics, aluminium, steel and steel reinforcement. Eco-reinforcement is a trademark for responsibly sourced reinforcing steel. It is a third-party certification scheme developed by the reinforcing steel industry to comply with BRE's Framework Standard Responsible Sourcing of Construction Products (BES 6001).

Using construction materials that have been recycled and are low impact offers a number of environmental and business benefits.

- Demonstrating performance against corporate and sustainability policies.
- Reducing material costs (where locally reprocessed demolition materials are cheaper than virgin materials).
- Supporting sustainability goals, to meet the requirements of clients and planning authorities.
- Providing a competitive edge through differentiation.
- Complementing other aspects of sustainable design.
- Conserving finite natural resources by reducing the demand for raw materials.
- Conserving energy and water, as recycled materials require less processing than extracting raw materials.
- Reducing air and water pollution, since manufacturing from recycled materials is generally a cleaner process and uses less energy.
- Reducing the amount of material that would otherwise go to landfill.

Designers should follow the principles of the circular economy, incorporating opportunities to reuse or recycle materials at the end of a product's life (for example, designing for dismantling and reuse of building components).

8.5.1 Demolition and refurbishment information data sheets (DRIDS)

The demolition and refurbishment information data sheets (DRIDS) have been developed to help identify waste streams and explain how they can be reused and recycled. They have been developed by the demolition industry, by giving consideration to knowledge and information available on materials they currently manage, or will be required to manage, in the future.

The design and construction of buildings, structures and infrastructures are now governed by tighter regulation and legislation. Therefore, the use of DRIDS allows users to be better informed about the types of materials and products they will encounter and how best to maximise environmental and economic gain.

There are 13 **DRIDS groups** plus a miscellaneous group, each with their own distinct shape, colour and code, to ease recognition. Each group contains a number of DRIDS, depending on their material or product make-up. For example, the wood group includes DRIDS that are generally one product, such as plywood, chipboard or dimensional timbers. However, wood furniture or wood framed glass panel systems may be in the composites group.

The colours have been chosen to reflect those used by BRE, WRAP, Institution of Civil Engineers (ICE) and Zero Waste Scotland. The groups include the following:

Composite (C1)

Electrical

Flooring (F1, F2)

Hazardous (H2)

Inert (I1, I2, I3)

Insulation (Z1)

Miscellaneous

Metal (M1, M2, M5)

Organic matter

Packaging

Plasterboard (G1)

Plastic (P1)

Wood (W1, W2, W3)



DRIDS are a useful resource for the wider construction industry.



For further information on DRIDS visit the National Federation of Demolition Contractors website or download the latest app.